



# Facilitator Guide

Sector

**Green Jobs**

Sub Sector

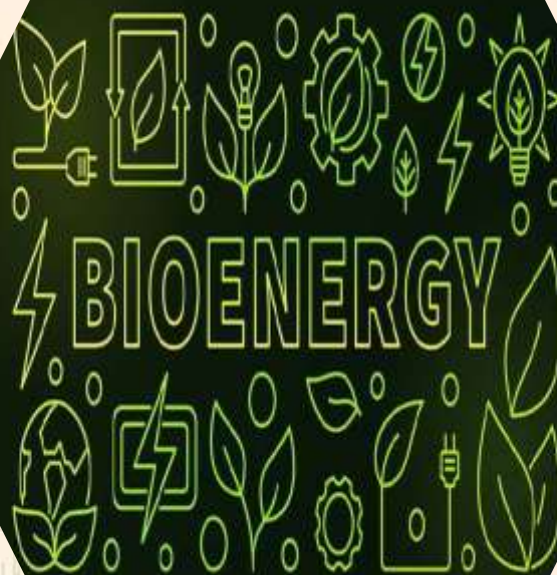
**Renewable Energy**

Occupation

**Entrepreneur**

Reference Id

**SGJ/Q4102, Version 1.0 NSQF Level - 4**



**Bio-energy  
Microentrepreneur**



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“

Skilling is building a better India.  
If we have to move India towards  
development then Skill Development  
should be our mission.

”

**Shri Narendra Modi**  
Prime Minister of India





## Certificate

### COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

**SKILL COUNCIL FOR GREEN JOBS**

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### SKILLING CONTENT: FACILITATOR GUIDE

Complying to National Occupational Standards of  
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The development of this Facilitator Guide on **Bio-Energy Micro Entrepreneur** (QP Code: SGJ/Q4102) has been made possible through the collective efforts of industry experts, academicians, and practitioners in the field of bio-energy. We extend our sincere gratitude to all individuals and organizations who have contributed their valuable insights, expertise, and time in shaping this resource.

Special thanks to the subject matter experts, training providers, and stakeholders from the biomass energy, biofuels, and biogas industries who provided real-world perspectives, ensuring the relevance and practical applicability of the content. Their contributions have enriched the learning experience for aspiring entrepreneurs in the bio-energy sector.

We also acknowledge the efforts of content developers, instructional designers, and reviewers who have meticulously structured and refined this guide to make it a comprehensive and user-friendly resource for facilitators.

Finally, we extend our gratitude to all trainers, learners, and industry partners who will utilize this guide to enhance skills, promote entrepreneurship, and contribute to the growth of the bio-energy sector. Your dedication to fostering knowledge and sustainable business ventures is invaluable in supporting India's clean energy transition.

Wishing all facilitators and learners success in their journey towards empowering bio-energy entrepreneurship.



## About this Book

This Facilitator Guide is designed to support trainers in delivering effective and engaging training for the **Bio-Energy Micro Entrepreneur** Qualification (QP Code: SGJ/Q4102). It serves as a comprehensive resource to assist facilitators in equipping learners with the necessary knowledge and practical skills to establish and manage enterprises in the bio-energy sector.

The Facilitator Guide on **Bio-Energy Micro Entrepreneur** is structured to provide clear instructions, teaching methodologies, and activity-based learning strategies. It covers essential topics including biomass energy production, financial management, marketing strategies, regulatory compliance, and the operational aspects of biomass pellet and biogas plant management. By following this guide, facilitators can ensure that training sessions are aligned with industry standards and the qualification's objectives.

This guide is tailored for training professionals, instructors, and subject matter experts who are committed to developing a skilled workforce in the bio-energy sector. It offers detailed lesson plans, practical exercises, case studies, and assessment methods to enhance the learning experience.

The Facilitator Guide provides a structured and interactive approach to delivering knowledge on sustainable bio-energy solutions. With this support, trainers can empower learners to contribute to the growth of the renewable energy sector while exploring profitable entrepreneurial opportunities. We hope this guide serves as an invaluable resource in shaping competent and confident bio-energy micro-entrepreneurs.

## Symbols Used



Key Learning  
Outcomes



Steps



Exercise



Activity



Notes



Unit  
Objectives

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## Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed:



[Employability skills workbook](#)





# 1. Entrepreneurship and Basics of Biomass Energy

Unit 1.1 - Fundamentals of Biofuels and Biomass Energy.

Unit 1.2 - Entrepreneurship in the Biomass Energy Sector

Unit 1.3 - Managing a Biomass-Based Enterprise

Unit 1.4 - Biomass Resource Management and Procurement



## Key Learning Outcomes

**At the end of this module, you will be able to:**

1. Understand Biofuels as an Energy Source
2. Define biofuels and their role in sustainable energy production.
3. Differentiate between solid, liquid, and gaseous biofuels with examples.
4. Explain the concept of entrepreneurship and its significance in the bio-energy sector.
5. Describe the key steps of the entrepreneurial process, from idea generation to investment harvesting.
6. Differentiate between a business model and a strategy in the context of biomass-based businesses.
7. Identify management challenges (financial, human resources, product/service development, operations) in MSMEs and startups.
8. Explain the significance of agro-residue waste as a resource and its management for energy applications.
9. Discuss various biomass energy conversion technologies, including biomass pellets and biogas.
10. Explain biomass procurement systems and the challenges involved.
11. Describe the cropping pattern, harvesting pattern, storage, and sale of biomass feedstock.
12. Demonstrate methods to assess biomass resource potential, pricing, and supply chain challenges.
13. Demonstrate how to conduct field surveys with farmers, aggregators, and FPOs to assess agricultural residue availability.

## UNIT 1.1: Fundamentals of Biofuels and Biomass Energy

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain the concept of biofuels as a renewable energy source.
2. Identify different types of biofuels (solid, liquid, and gaseous).
3. Describe the role of biomass in the global energy transition.

### Ask

Ask the participants to introduce. Then introduce yourself to the participants. Tell the learners how this presentation will benefit them. Since mature learners are more interested in a subject if they know how or why it is important to them.

### Say

Welcome to this session on Fundamentals of Biofuels and Biomass Energy

- "Today, we will discuss biofuels and biomass energy as renewable energy sources, their types, and their importance in the global energy transition."
- "Let's begin by understanding what biofuels are and why they are gaining importance in the renewable energy sector."

### Ask

- "What do you already know about biofuels?"
- "Can you name some sources of biofuels?"

### Explain

- Define biofuels as fuels derived from organic materials.
- Describe how biofuels are renewable and how they can reduce greenhouse gas emissions.
- Provide examples of common biofuels like bioethanol and biodiesel.
- Discuss their applications in transportation, heating, and electricity generation.
- Define biofuels as fuels derived from organic materials.

- Describe how biofuels are renewable and how they can reduce greenhouse gas emissions.
- Provide examples of common biofuels like bioethanol and biodiesel.
- Discuss their applications in transportation, heating, and electricity generation.
- Describe biomass as organic material from plants, animals, and waste.
- Discuss how biomass is converted to energy through combustion, biogas production, and gasification.
- Explain its contribution to reducing waste and lowering emissions.

## Notes for Facilitation

### Facilitator Tips:

- Provide local examples of biofuel use.
- Emphasize the benefits of biofuels over fossil fuels.
- Display images of solid biofuel samples like wood, pellets, and briquettes.
- Use visual aids to explain liquid biofuel production and usage.
- Share case studies on biogas plants or community biofuel projects.
- Display images of solid biofuel samples like wood, pellets, and briquettes.
- Use visual aids to explain liquid biofuel production and usage.
- Share case studies on biogas plants or community biofuel projects.

## Explain

Explain Why Is It Important to Sort and Understand Waste Streams?

You might think it's okay to just throw all the waste together in one big pile, but that's not a good idea. Each kind of waste is different, and they all need to be handled in different ways.

By sorting waste properly, we can recycle more efficiently, reduce waste, and follow the rules that help protect the environment. Understanding what type of waste, we have helps us know how to deal with it better, and it helps save money and resources.

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Group activity – Divide learners into three groups to research and present on one type of biofuel..
- Ask the group to discuss and write down the answers.
- Give the groups 15 minutes for the activity.

## Summarize

Summarize the topics covered in the module.

Reiterate the key concepts: definition of biofuels, production processes, applications, and benefits.

Recap how biomass energy supports the global energy transition, reduces emissions, and creates economic opportunities.

You may ask the learners to recall what all has been taught.



## UNIT 1.2: Entrepreneurship in the Biomass Energy Sector

### Unit Objectives

At the end of this unit, you will be able to:

1. Understand the Concept of entrepreneurship development
2. Explain the Steps of the entrepreneurial process
3. Discuss Business models vs. business strategies in the biomass industry.

### Say

Welcome to this session on Entrepreneurship in the Biomass Energy Sector

- Energy needs are increasing globally, and biomass energy offers a clean and sustainable alternative.
- Entrepreneurs play a critical role in reducing carbon emissions, generating jobs, and supporting rural economies.

### Ask

Ask the participants

- Can you name any examples of biomass energy sources in your region?
- Have you seen or heard of biomass-based products like biogas, pellets, or biofuels?

### Explain

- Provide examples of biomass ventures such as biogas plants, ethanol production, and biomass power generation.
- Explain how biomass energy reduces waste and supports sustainable development.

### Notes for Facilitation

- Share success stories of local biomass entrepreneurs.
- Discuss government support for biomass initiatives in your country.

### Say

- Entrepreneurship is about identifying opportunities, managing risks, and creating value.
- In the biomass sector, entrepreneurs convert organic waste into useful energy products.

## Ask

Ask the participants

- What skills do you think are necessary to become a successful biomass entrepreneur?
- What challenges might biomass entrepreneurs face?

## Explain

- Discuss the role of innovation and problem-solving in biomass entrepreneurship.
- Explain the social, economic, and environmental impact of biomass businesses.

## Notes for Facilitation

- Provide examples of different types of biomasses and their applications.
- Encourage learners to brainstorm potential biomass business ideas.

## Say

The entrepreneurial journey involves clear steps from identifying opportunities to growing a business.

## Explain

Present the seven steps of the entrepreneurial process with practical examples:

1. Identifying and Evaluating the Business Opportunity
2. Developing a Business Plan
3. Acquiring Financial and Non-Financial Resources
4. Setting Up the Business and Ensuring Compliance
5. Managing Production and Operations
6. Marketing and Commercialization
7. Monitoring, Evaluation, and Scaling Up

## Demonstrate

- Provide a case study of a biomass entrepreneur who followed these steps.
- Ask learners to analyze the case and identify which steps were critical to the entrepreneur's success.

## Team Activity



- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Form groups and assign each group a step in the entrepreneurial process.
- Ask each group to present how they would approach that step if they were starting a biomass business.
- Give the groups 5 minutes for the activity

## Say



- A business model defines how a company creates, delivers, and captures value.
- A business strategy outlines how the company competes and grows in the market.

## Ask



Ask the participants

- Which business model do you think is most viable in your region?
- How can a biomass company stand out in a competitive market?

## Explain



Compare different business models in the biomass industry, such as:

- Biomass-to-Energy Model
- Biofuel Production Model
- Biomass Waste Management Model
- Pellet and Briquette Manufacturing Model
- Discuss business strategies including cost leadership, differentiation, niche market, innovation, and sustainability.

## Notes for Facilitation



- Show visual aids comparing business models and strategies.
- Encourage learners to brainstorm a biomass business model and create a basic strategy for it.

## Summarize



Summarize the topics covered in the module.

Key concepts covered: Entrepreneurship development, the entrepreneurial process, business models, and strategies.

You may ask the learners to recall what all has been taught.

## UNIT 1.3: Managing a Biomass-Based Enterprise

### Unit Objectives

At the end of this unit, you will be able to:

1. Gain knowledge on management challenges (financial, human resources, product/service development, operations) in managing an emerging business or MSMEs

### Ask

Ask the participants

- What do you think are the key management challenges in running a biomass enterprise?
- Can you think of any successful biomass businesses in your region?

### Say

Welcome learners and introduce the session by discussing the importance of management in biomass enterprises.

Financial management is critical for the success of biomass enterprises. Securing funding, managing costs, and ensuring consistent cash flow are essential.

Effective operations management ensures smooth business activities and quality production.

### Explain

Highlight the key management aspects covered in this unit: financial management, human resource management, product/service development, and operational management.

- Provide examples of real-world biomass enterprises.
- Explain how biomass enterprises contribute to sustainable energy and rural development.
- Discuss various funding options such as government grants, subsidies, private investors, and loans.
- Explain cost management strategies, including resource optimization and cost tracking.
- Discuss various funding options such as government grants, subsidies, private investors, and loans.
- Explain cost management strategies, including resource optimization and cost tracking.
- Discuss various funding options such as government grants, subsidies, private investors, and loans.
- Explain cost management strategies, including resource optimization and cost tracking.
- Discuss supply chain management, infrastructure requirements, and compliance with environmental regulations.

## Demonstrate

- Provide a sample cash flow analysis for a biomass business.
- Guide learners in creating a simple financial budget.
- Show a sample supply chain diagram for a biomass enterprise.

## Notes for Facilitation

Emphasize the role of innovation and sustainability in biomass enterprise management.

- Encourage learners to share experiences from previous workplaces.
- Emphasize the importance of maintaining a safe and inclusive work environment.
- Provide insights into optimizing logistics and improving production efficiency.

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to form groups to design a budget plan for a biomass enterprise using hypothetical data.
- Ask the group to discuss.

Give the groups 5 minutes for the activity.

## Role Play

Educate them on how to ask the previously explained pertinent questions.

Then Conduct a role-play exercise where one learner acts as an HR manager addressing employee concerns.

## Summarize

Summarize the topics covered in the module.

Summarize the key management challenges and solutions discussed in this unit.

You may ask the learners to recall what all has been taught.

## UNIT 1.4: Biomass Resource Management and Procurement

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain biomass procurement system and its challenges.
2. Explain the importance of Agro-residue waste as a resource and its management for end use applications.

### Say

Welcome to unit Biomass Resource Management and Procurement. Today, we'll explore how biomass, a vital renewable resource, can be effectively managed and procured for sustainable energy production. We'll dive into the procurement process, tackle its challenges, and highlight the untapped potential of agro-residue waste.

### Ask

Ask the participants

- What do you think are the biggest hurdles in sourcing biomass for energy projects?
- How many of you have seen farmers burning crop residues? Why do you think they do it?
- What qualities would you look for in a biomass supplier?
- How might agro-residue waste benefit your community or business?

### Explain

#### ▪ Biomass Procurement System and Its Challenges

Key Points:

Procurement involves sourcing sustainable, cost-effective, high-quality biomass.

Steps: Choosing suppliers, ensuring quality, managing costs, logistics, legal compliance, innovation, and stakeholder collaboration.

Challenges: Seasonal availability, transportation difficulties, low farmer incentives, and lack of automation.

Solutions: Storage facilities, compressing biomass, increasing farmer incentives, and adopting technology.

#### ▪ Importance of Agro-Residue Waste as a Resource

### Key Points:

- Types: Primary (e.g., rice straw) and secondary residues (e.g., bagasse).
- Benefits: Renewable energy, pollution reduction, soil health, livestock feed, industrial use, and circular economy.
- Management Strategies: Collection/storage, processing, technology integration, policy support, and farmer awareness.
- Case Study: Highlight Husk Power Systems (HPS) in India as a success story.

## Demonstrate

Demonstrating Biomass Quantity, Consistency, Quality, Cost, and Availability

### Steps:

1. **Define Biomass Needs:** Present a hypothetical energy demand (e.g., 1 MW plant requiring 10 tons/day). Use the formula:  $\text{Biomass Required (tons)} = \frac{\text{Energy Demand (MJ)}}{\text{Calorific Value (MJ/kg)} \times \text{Conversion Efficiency}}$ .
2. **Feedstock Options:** Display samples or images of rice husk, bagasse, and wood chips; discuss their pros/cons.
3. **Quality Testing:** Simulate a quality check using a moisture meter (or describe its use if unavailable).
4. **Availability Mapping:** Show a simple GIS map (pre-prepared) of a biomass-rich area and explain how to identify “hotspots.”

## Notes for Facilitation

- **Adaptability:** Tailor examples to the local context (e.g., use regional crops like maize or bamboo).
- **Engagement:** Use humor or anecdotes (e.g., “Ever tried hauling straw in a rainstorm?”) to keep energy high.
- **Resources:** Prepare handouts with key terms, formulas, and case studies for reference.
- **Evaluation:** Optionally, distribute a short quiz or feedback form to gauge understanding.

## Team Activity

- Divide the class into five to six groups of four participants each.
- Ask the groups to Plan a supply chain for a 5 MW biomass plant using agro-residue waste in a rural area.
- Consider: Supplier selection, logistics, storage, cost management, and farmer incentives.
- Give the groups 5 minutes for the activity. Teams present their plans in 5 minutes.

## Summarize

Summarize the topics covered in the module.

Recap key points:

- Biomass procurement requires careful planning and stakeholder collaboration.
- Agro-residue waste is a versatile resource with economic and environmental benefits.
- Challenges like seasonality and logistics can be mitigated with innovation and policy support.

You may ask the learners to recall what all has been taught.







## 2. Manage financial budget and developing business plans

Unit 2.1– Fundamentals of Financial Management for Biomass Enterprises

Unit 2.2 – Financing Strategies for Biomass Ventures

Unit 2.3 – Business Development and Innovation in Biomass Energy

Unit 2.4 – Operations and Technology in Biomass Pellet and Biogas Plants

Unit 2.5 – Supply Chain, Storage, and Budgeting for Biomass Units



## Key Learning Outcomes

**At the end of this module, you will be able to:**

1. Define the fundamental concepts and components of financial statements.
2. Explain the purpose, structure, and interpretation of balance sheets, profit and loss statements, and cash flow statements.
3. Apply and calculate the break-even point to assess business viability.
4. Identify different sources of financing and evaluate their advantages and risks.
5. Assess key financial management aspects, including cash flow management and financial planning for biomass and biogas businesses.
6. Explain the concept of entrepreneurship development and its importance in the bio-energy sector.
7. Describe the key stages of the entrepreneurial process, from ideation to implementation and growth.
8. Explain business development activities, including market research, customer engagement, and strategic planning.
9. Demonstrate the entire operational process of a biomass pellet plant, covering raw material collection, unloading, production, storage, and transportation.
10. Identify, operate, and maintain tools, equipment, and machinery used in biomass pellet plants and biogas units.
11. Evaluate different biomass energy technologies for pellet production and biogas generation.
12. Describe the significance of agro-residue waste as a resource and its management for end-use applications.
13. Demonstrate various storage options for biomass pellets and evaluate factors influencing their shelf life and quality.
14. Understand different storage solutions for biogas and their role in ensuring supply chain efficiency.
15. Develop a sustainable and feasible business plan for a biomass pellet or biogas unit.
16. Understand the financial implications and budgetary considerations for setting up and operating a biomass production plant.
17. Understand the end-to-end operations of a biomass pellet plant, including raw material sourcing, transportation, storage, production processes, and logistics.
18. Demonstrate the process of setting up a biomass pellet or biogas unit, including assessment of location, catchment area, demand, and supply analysis.

## UNIT 2.1: Fundamentals of Financial Management for Biomass Enterprises

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Define the fundamental concepts and components of financial statements.
2. Explain the purpose, structure, and interpretation of balance sheets, profit and loss statements, and cash flow statements.
3. Apply and calculate the break-even point to assess business viability.

### Say

Welcome to the session on Fundamentals of Financial Management for Biomass Enterprises. The financial success of any biomass enterprise is directly influenced by effective financial management. Understanding financial statements and break-even analysis is crucial for making informed decisions.

### Ask

Ask the participants

- Why do you think financial management is essential for a biomass business?
- What financial challenges do you anticipate in a biomass enterprise?

### Explain

- Provide an overview of financial management principles.
- Discuss how financial decisions impact business sustainability and profitability.

### Notes for Facilitation

Use real-world biomass enterprise examples to explain the importance of financial management.

Emphasize the role of government subsidies, grants, and incentives in financial planning.

### Say

Financial statements are structured records that reflect a company's financial health and guide decision-making. Let's explore the fundamental concepts

## Ask

Ask the participants

- What do you think is the difference between accrual accounting and cash accounting?
- Can you think of different revenue sources for a biomass enterprise?

## Explain

- Explain accrual vs. cash accounting.
- Discuss revenue recognition and cost structures using biomass industry examples.
- Illustrate asset valuation methods like FIFO and Weighted Average Costing.
- How do you think a balance sheet helps business owners make financial decisions?
- What can a profit and loss statement reveal about a biomass company's financial health?
- Break-even analysis is a valuable tool for understanding the minimum sales needed to cover costs.

## Notes for Facilitation

- Provide sample financial statements from a biomass enterprise.
- Use visual aids to explain how different transactions are recorded.
- Provide sample financial statements from a biomass enterprise.
- Use visual aids to explain how different transactions are recorded.
- Provide sample financial statements from a biomass enterprise.
- Use visual aids to explain how different transactions are recorded.

## Demonstrate

Walk through a sample cash flow statement to illustrate how to track cash inflows and outflows.

Solve a sample break-even point problem using realistic biomass enterprise data.

## Do

You may Show a video.

### VIDEO

Topic: Break-Even Analysis with Easy Explanation

By: gurushiya

Link: <https://www.youtube.com/watch?v=uKeopB8acGQ>

Duration: 10 m 01 sec



Scan Qr to watch video

## Role Play



Educate them on how to ask the previously explained pertinent questions.  
Ask learners to play the role of a financial analyst evaluating a biomass enterprise's financial statements.

## Summarize



Summarize the topics covered in the module. Like  
Break-even analysis is essential for determining financial feasibility and guiding pricing and cost management decisions.

Emphasize the importance of accurate financial reporting.

Reinforce the significance of financial analysis in ensuring business sustainability.

You may ask the learners to recall what all has been taught.

## UNIT 2.2: Financing Strategies for Biomass Ventures

### Unit Objectives

At the end of this unit, you will be able to:

1. Explain the different stages of financing.
2. Define the characteristics of the two basic types of financing: debt and equity.
3. Describe the appropriateness of types of financing at various stages of an organization's life.
4. Identify the different sources of financing.
5. Identify the important issues in managing a firm's cash flows.

### Say

Welcome to Unit Financing Strategies for Biomass Ventures! Today, we're diving into the financial backbone of biomass and bioenergy projects. These ventures play a key role in renewable energy, but they require significant funding to succeed. We'll explore the stages of financing, the types of funding available, and how to manage cash flow effectively.

### Ask

Ask the participants

- Why do you think financing is critical for biomass ventures?"
- Have any of you been involved in a project that needed funding? What challenges did you face?

### Explain

There are six stages:

1. **Idea and Research (Pre-Seed):** Funding for studies and small tests—often from personal savings or grants.
2. **Proof of Concept (Seed):** Building a pilot—angel investors or government innovation funds step in.
3. **Early-Stage Deployment (Startup):** Small commercial setup—venture capital or bank loans help.
4. **Expansion (Growth):** Scaling up—debt financing or green bonds are common.
5. **Full-Scale Operations (Maturity):** Optimizing and diversifying—corporate loans or stock market funding.
6. **Decline/Renewal:** Reinvesting or exiting—profits or strategic partnerships."

## Demonstrate

Show Figure 9 (Stages of Business Financing) from the text on a slide or handout. Point out how funding needs escalate and diversify over time

## Notes for Facilitation

- Use simple examples (e.g., "Imagine testing a small biomass stove vs. building a large plant").
- Encourage creativity in the team activity—let them think like entrepreneurs.

## Say

Now let's look at the two main ways to get money: debt and equity. Each has pros and cons.

## Ask

Ask the participants

- What's the difference between borrowing money and selling a piece of your company?"
- Which do you think is riskier for a business owner?

## Explain

**Debt Financing:** You borrow money and repay it with interest. You keep control, but you must pay back no matter what. Examples include bank loans or green bonds.

**Equity Financing:** You sell ownership shares to investors. No repayment, but you share control and profits. Think angel investors or venture capital.

## Demonstrate

Show a video

### VIDEO

Topic: Sources of finance

By: after the bell

Link: <https://www.youtube.com/watch?v=Jj0LJQlee14>

Duration: 12 m 29 sec



Scan Qr to watch video



## Notes for Facilitation

- Emphasize real-world trade-offs (e.g., "Debt means pressure to repay; equity means less control").
- Keep the role play light and fun—encourage negotiation creativity.
- Connect to learners' lives: "If you started a business, would you borrow or find a partner?"
- Reinforce that biomass projects often mix both types.
- Mention local examples if possible (e.g., regional grants).
- Keep it practical—learners should feel they could research these sources.
- Use relatable analogies (e.g., "It's like budgeting for a big party").
- Stress proactive planning—learners should feel empowered.

## Team Activity

- Divide the class into five to six groups of four participants each.
  - Give blank paper and pens to each group.
  - Assign each group one financing stage.
  - Give them 10 minutes to brainstorm and present: Two funding sources for their stage. One challenge a biomass company might face at that stage.
- Debrief: Discuss how needs shift across stages.

## Summarize

Summarize the topics covered in the module. For eg.

We covered the financing stages—from idea to maturity—and how debt and equity fit each one. We explored sources like loans, investors, and grants, and tackled cash flow challenges like seasonal costs. Biomass ventures need smart financing to thrive."

Ask

"What's one thing you learned that surprised you?"

"How could you use this in a real project?"

## UNIT 2.3: Business Development and Innovation in Biomass Energy

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain business development tasks and activities
2. Understand how the sustainable growth model is derived and apply it to determine financing requirements for a new venture.
3. Explain the characteristics of a good innovation to develop a business plan.

### Say

Welcome to Unit: Business Development and Innovation in Biomass Energy! Biomass energy is a growing field with huge potential for businesses that want to make a profit while helping the planet. Today, we'll explore how to build a biomass business, plan its growth sustainably, and use innovation to stand out in the market.

### Ask

- Why do you think biomass energy is an exciting business opportunity?
- What challenges might someone face when starting a new energy business?
- What do you think is the first thing a biomass business needs to do to get started?"
- How might partnerships help a new energy company?
- What happens if a business grows too fast without enough cash?
- How could a biomass company use its profits to expand?
- What makes a new product or idea successful in your opinion?"
- "Why might sustainability matter for a biomass innovation?
- Which part of today's session clicked most for you?"
- How could you use one of these ideas in a biomass project?

### Demonstrate

Show a flowchart on a slide or board: Market Research → Technology → Legal → Finance → Partnerships → Marketing → Execution → Innovation. Add a biomass example, like "Researching demand for biogas in rural areas."

### Role Play

Scenario: One learner is a biomass entrepreneur, another a bank manager. The entrepreneur pitches a 20% growth plan and asks for a loan to cover the 5.6% gap (5 minutes). Switch roles. Debrief: "What financing option felt most convincing? Why?"

## Explain

Business development in biomass involves eight key tasks:

1. **Market Research:** Understand demand, customers, and competition.
2. **Technology Selection:** Pick the best method—like combustion or gasification—for your biomass.
3. **Legal Compliance:** Get permits and follow environmental rules.
4. **Financial Planning:** Estimate costs and find funding.
5. **Partnerships:** Work with farmers or energy buyers for supply and sales.
6. **Marketing:** Promote your biomass solution to attract customers.
7. **Project Execution:** Build and run the plant efficiently.
8. **Innovation:** Keep improving and stay sustainable.

These steps ensure your business is built on a strong foundation.

**Sustainable Growth Rate (SGR)** tells us how much a business can grow using its own profits, without extra loans or investors. The formula is:

$$\text{SGR} = \text{ROE} \times (1 - \text{Dividend Payout Ratio})$$

- **ROE** is profit divided by equity—how well you use your money.
- **Dividend Payout Ratio** is how much profit you give to shareholders vs. keep. For example, if ROE is 15% and you keep 80% of profits,  $\text{SGR} = 15\% \times 0.8 = 12\%$ . If you want to grow faster—say 18%—you'll need extra financing like loans or grants. Biomass firms use this to plan smart growth.

**A good innovation has 10 key traits:**

Market Relevance: Meets a real need, like cheap renewable energy.

Scalability: Can grow from small to big.

Cost-Effectiveness: Saves money for you and customers.

Sustainability: Cuts emissions and uses resources wisely.

Regulatory Fit: Follows laws and gets incentives.

IP Protection: Patents keep it unique.

Adaptability: Evolves with new tech or markets.

Commercial Fit: Ties into a solid business plan.

Risk Management: Plans for challenges.

Consumer Buy-In: People trust and want it.

These traits turn an idea into a profitable biomass business.

## Summarize

Summarize the topics covered in the module. For ex.

Today, We've covered how to build a biomass business with tasks like market research and partnerships. We used the sustainable growth model to plan financing—balancing profits and external funds. And we defined a good innovation—market-driven, scalable, and sustainable. You're now ready to develop a biomass venture that grows smart and innovates well!

## UNIT 2.4: Operations and Technology in Biomass Pellet and Biogas Plants

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Discuss specific requirement: electric load, temperature, pressure, feeding cycle and capacity, manpower requirement at each unit.
2. Discuss dos and don'ts while assembling components of machines.
3. Discuss requirements of end product so specific procedures can be followed accordingly.

### Say

Welcome to Unit: Operations and Technology in Biomass Pellet and Biogas Plants! Biomass energy is a game-changer in the shift to renewables, and today we'll dive into how biomass pellet and biogas plants work. We'll cover the key operations, the tools and machines that make it happen, and how to pick the best technology for the job.

### Ask

Ask the participants

- Why do you think biomass pellets or biogas could replace fossil fuels?
- What might be some challenges in running a plant that turns waste into energy?
- What do you think happens to wood chips or straw before they become pellets?
- Why might storage matter for pellet quality?
- What kind of machine might grind wood into tiny bits?
- Why would a biogas plant need something to stir waste?
- What might make one pellet machine better than another?"
- How could the type of waste affect a biogas system choice?

### Explain

**A biomass pellet plant has four main operations:**

1. Unloading Raw Materials: Biomass like sawdust or crop residues arrives, gets inspected, weighed, and moved to storage.
2. Production Process: This includes drying (to reduce moisture), grinding (to make small particles), pelletizing (compressing into pellets), and cooling (to harden them).
3. Loading Pellets: Pellets are packed into bags or bulk containers for shipping.
4. Storage: Pellets are kept in silos or warehouses to stay dry and intact. Each step uses specific machines to keep things efficient and high-quality.

**Pellet Plants** use tools like moisture meters (to check wetness) and wrenches (for fixes), plus big machines:

- **Hammer Mills:** Grind biomass into fine particles.
- **Pellet Mills:** Shape it into pellets.
- **Rotary Dryers:** Remove moisture.
- **Conveyor Belts:** Move stuff around.

**Biogas Units** use tools like pH meters (to monitor digestion) and gas leak detectors (for safety), plus machines:

- **Digesters:** Turn waste into gas.
- **Agitators:** Stir the mix for better gas output.
- **Purification Systems:** Clean the biogas.
- **Generators:** Make electricity.

These tools keep everything running smoothly and safely.

**For Pellets:** Pick tech based on:

- **Raw Material:** Wood needs ring die mills; straw might use flat die.
- **Scale:** Small plants use flat die; big ones use ring die or integrated systems.
- **Efficiency:** Look for low energy use and high output.
- Proven options: Ring Die Mills (big scale), Flat Die Mills (small scale).

**For Biogas:** Consider:

- **Feedstock:** Manure suits fixed-dome; food waste fits CSTR.
- **Size:** Household uses fixed-dome; industry uses CSTR or UASB.
- **Yield:** Stirred systems boost gas.
- Proven options: Fixed-Dome (small), CSTR (large).

Check Table 7—it compares pellets (dry, solid fuel) vs. biogas (wet, gas fuel). The right tech matches your needs and resources.

## Demonstrate

Show Figure 10 (Unloading Process) and Figure 11 (Pellet Manufacturing Process) on slides or handouts. Walk through the flow: “Truck unloads sawdust → dryer reduces moisture → pellet mill shapes it → cooler hardens it → bags or silos.”

Use a simple prop (e.g., a handful of wood chips and a small pellet) to show the transformation.

Show images from Figures 12-34 (e.g., Hammer Mill, Biogas Digester) on slides. Point out one tool and one machine for each plant type, explaining their jobs (e.g., “This pellet mill squeezes sawdust into pellets”).

If possible, bring a small tool (like a moisture meter) to pass around.

Draw a quick decision tree on a board:

- Pellets: Wood? → Ring Die. Small scale? → Flat Die.
- Biogas: Manure? → Fixed-Dome. Big plant? → CSTR.

Highlight one tech per type (e.g., “Ring Die Mill makes dense pellets fast”).

## Notes for Facilitation

- Use visuals or props to make it tangible—learners should “see” the process.
- Keep explanations practical (e.g., “Drying stops pellets from crumbling”)
- Focus on a few key items—don’t overwhelm with the full list.
- Make it interactive—let learners guess machine purposes before explaining.
- Simplify factors—focus on feedstock and scale as starting points.
- Encourage debate in the activity—let learners defend their choices.

## Team Activity

- Divide the class into five to six groups of four participants each.  
**Activity:** “Pellet Plant Walkthrough.” Divide learners into four groups, each assigned an operation (Unloading, Production, Loading, Storage). In 10 minutes, they:  
List two key steps in their operation.  
Name one machine used and its role.  
Present their findings.  
Debrief: Discuss how the operations link together like a chain.

## Summarize

Summarize the topics covered in the module.

- We walked through a pellet plant’s operations—unloading, producing, loading, and storing—and saw the tools and machines that make it work, like pellet mills and digesters. We also learned how to pick proven tech for pellets and biogas based on feedstock, scale, and efficiency. You’re now ready to run or design a biomass energy system!

## UNIT 2.5: Supply Chain, Storage, and Budgeting for Biomass Units

### Unit Objectives

At the end of this unit, you will be able to:

1. Demonstrate value-chain covering demand & supply of biomass pellet and biogas considering a suitable catchment area to set up unit.
2. Show them storage options of Biomass pellets and its shelf life
3. Show them storage option for biogas
4. Demonstrate budgetary cost for setting up a Pellet plant and small biogas production unit.

### Say

Welcome to Unit 2.5: Supply Chain, Storage, and Budgeting for Biomass Units! Biomass pellets and biogas are renewable energy stars, but making them work means planning the supply chain, storing them right, and budgeting smartly. Today, we'll explore how to get raw materials to market, keep products fresh, and figure out the costs to start these ventures.

### Ask

Ask the participants

- Why do you think location matters for a biomass plant?
- What might happen if pellets or biogas aren't stored properly?
- Who might want to buy biomass pellets or biogas?
- What could make it hard to get enough raw materials?
- What might ruin a pile of pellets?
- How long do you think pellets can last if stored well?
- Why might storing gas be harder than storing pellets?
- What could go wrong if biogas leaks?

### Explain

The **value chain** covers demand, supply, and location:

- **Demand:** Industries (e.g., cement plants), power plants, households (cooking), farms (fertilizer), and businesses (heating) want pellets and biogas.
- **Supply:** Raw materials come from agricultural waste (straw), forestry waste (sawdust), animal waste (manure), and energy crops (switchgrass).
- **Catchment Area:** Pick a spot near raw materials and markets, with good roads and permits.

Key steps:

1. **Procurement:** Get materials from farmers or sawmills.
2. **Production:** Turn them into pellets or gas.
3. **Storage/Distribution:** Keep them safe and deliver them.

4. **Market Linkages:** Sell to steady buyers.

A good chain keeps everything flowing smoothly.

**"Storage Options:**

- **Indoor:**

- **Silos:** Airtight, great for bulk.
- **Bunkers:** Protected, need ventilation.
- **Warehouses:** Controlled, ideal for big stocks.

- **Outdoor:**

- **Tarps:** Cheap, but risky for moisture.
- **Sheds:** Better cover, need elevation.

- **Bulk:**

- **Vertical Silos:** Automated, space-saving.
- **Domes:** Cost-effective for huge amounts.

**Shelf Life:** 6 months to 2 years, depending on:

- **Moisture:** Keep below 10%—wet pellets crumble.
- **Temperature/Humidity:** Cool and dry stops mold.
- **Pests:** Seal them tight.

**Best Practices:** Use airtight bags, ventilate, store off the ground, and use older stock first."

**"Storage Options:**

- **Low-Pressure (small scale):**

- **Floating Drum:** Moves with gas levels, simple.
- **Fixed Dome:** Durable, low pressure.
- **Balloon:** Cheap, portable, but fragile.

- **Medium-Pressure (commercial):**

- **Gas Holders:** Big tanks, steady supply.
- **Water-Sealed Tanks:** Leak-proof, needs water checks.

- **High-Pressure (industrial):**

- **Compressed (CBG):** Cylinders, easy to move.
- **Liquefied (LBG):** Super compact, costly.

**Key Factors:** Scale (small farm vs. factory), use (cooking vs. power), cost, climate, and safety. Methane's flammable, so no leaks allowed!"

**Pellet Plant (500-1000 kg/h):**

- **Capital Costs:** \$61,000-\$167,000 (land, machines like pellet mills, dryers).
- **Monthly Costs:** \$14,000-\$39,000 (raw materials, labour, power).
- **Break-even:** 12-24 months.

**Biogas Unit (10-50 m<sup>3</sup>/day):**

- **Capital Costs:** \$31,000-\$82,000 (digester, storage, optional generator).
- **Monthly Costs:** \$3,500-\$12,000 (waste, labour, maintenance).
- **Break-even:** 18-36 months.

See Tables 8-12—pellets cost more upfront but scale faster; biogas is cheaper but takes longer to pay off."

## Demonstrate

Show Table 12 (Comparison) on a slide. Highlight: "Pellet plant: \$61K-\$167K start, vs. Biogas: \$31K-\$82K. Monthly costs differ too.



## Notes for Facilitation

- Start with a relatable analogy (e.g., a kitchen) to make it approachable.
- Use “Ask” to get learners thinking about real-world implications.
- Prepare a slide or board with the four objectives as a roadmap.
- Use local examples if possible (e.g., nearby farms or industries).
- Keep it practical—focus on linking supply to demand.
- Bring a pellet sample (if available) to show texture and discuss moisture risks.
- Emphasize shelf life factors—learners should feel they can protect quality.
- Simplify numbers—focus on big-ticket items (e.g., pellet mill, digester).
- Connect costs to earlier sections (e.g., storage needs).

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.

**Activity:** “Pick Your Storage.” Groups choose a biogas use (e.g., home cooking, factory power). In 10 minutes:

Select a storage option.

Explain why it fits (e.g., cost, scale).

Share with the class.

Debrief: “Did your choice balance cost and safety?”

## Summarize

Summarize the topics covered in the module.

- Today, we mapped the value chain—supplying pellets and biogas from waste to users in a smart location. We explored storage: pellets need dry silos or sheds (6 months-2 years shelf life), biogas needs safe tanks or cylinders. And we budgeted: \$61K-\$167K for a pellet plant, \$31K-\$82K for biogas. You’re ready to plan a biomass business!"
- You may ask the learners to recall what all has been taught.





## 3. Manage sales, supply, and marketing of product

Unit 3.1– Marketing Strategies for Biomass Pellets and Biogas

Unit 3.2 – Sales Planning and Customer Engagement

Unit 3.3 – Market Research and Performance Assessment

Unit 3.4– Organic Farming Economics and Sales Models



## Key Learning Outcomes

### **At the end of this module, you will be able to:**

1. Explain various sales promotion tools to increase the demand for biomass pellets and biogas in industrial and domestic sectors.
2. Apply the 4Ps of Product Marketing (Product, Price, Promotion, Place) and the 4Ps of Service Marketing (People, Processes, Physical Evidence, Productivity) in developing a comprehensive marketing strategy.
3. Develop an estimate of sales for biogas or biomass pellets by analyzing industry and household consumption patterns.
4. Calculate revenue projections and assess business feasibility.
5. Describe the sales process and execute key sales steps including lead generation, pitching, negotiation, and closing deals.
6. Identify best practices to ensure efficient sales operations.
7. Apply customer satisfaction tracking techniques to gather feedback and improve service quality.
8. Evaluate opportunities for global marketing of biomass pellets and biogas.
9. Discuss the potential of MSMEs as global enterprises in the renewable energy sector.
10. Identify and apply criteria to assess marketing performance using sales metrics, market share, and customer satisfaction data.
11. Conduct primary and secondary research to identify target customers such as thermal power stations, industries, and households.
12. Create customer profiles and analyze their energy consumption patterns.
13. Calculate the cost of production for conventional versus organic farming systems.
14. Evaluate the benefits of organic farming in terms of reduced costs and environmental impact.
15. Estimate the profitability of organic farming through direct-to-consumer sales.
16. Explore strategies for maximizing profits by reducing intermediaries.
17. Assess the scope and role of e-commerce in promoting and selling organic products.
18. Identify procedures for selling organic produce in local markets or through organic huts.

## UNIT 3.1: Marketing Strategies for Biomass Pellets and Biogas

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Discuss the use of sales promotion tools to increase demand of biomass pellets or Biogas.
2. Review the 4Ps of product marketing (Product, Price, Promotion, Place) and 7Ps of service marketing (People, Processes, Physical Evidence, Productivity).
3. Evaluate and demonstrate the scope & way of ecommerce in selling the produce to the consumers

### Say

Welcome to this Unit.

- Today we will explore how effective marketing strategies can boost the demand for biomass pellets and biogas.”
- “We’ll look into sales promotion tools, the marketing mix, and how e-commerce can help businesses grow.

### Ask

Ask the participants

- How would you set competitive pricing for biomass products?
- What promotional channels would be most effective for biomass enterprises?
- How would you set competitive pricing for biomass products?
- What promotional channels would be most effective for biomass enterprises?

### Explain

- Provide an overview of why marketing strategies are essential for biomass and biogas enterprises.
- Emphasize the importance of educating consumers about the environmental and economic benefits.
- Define sales promotion and explain its role in generating quick market responses.
- Discuss the objectives of sales promotion for biomass pellets and biogas.
- Introduce the 4Ps of product marketing (Product, Price, Promotion, Place).

- Discuss how the 7Ps of service marketing apply to biomass and biogas services.
- Highlight the scope of e-commerce in expanding market reach for biomass products.
- Discuss different platforms, from company websites to third-party marketplaces.

## Demonstrate

- Provide examples of consumer-oriented, trade-oriented, and digital sales promotion tools.
- Show a short video of a successful biomass or biogas promotional campaign.
- Show a live example of an e-commerce platform selling biomass pellets or biogas.
- Walk through the process of listing a product, setting prices, and managing inventory.

## Notes for Facilitation

- Encourage participants to use real-world examples from their experience.
- Provide additional insights on government subsidies and certifications that influence product positioning.

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Give them a task asking them to Develop a simple digital marketing plan to promote their products online.
- Ask the group to discuss and write down the answers.
- Give the groups 15 minutes for the activity.

## Role Play

Educate them on how to ask the previously explained pertinent questions.

- Simulate a scenario where Participants role-play as business owners and customers.
- Create scenarios where they must apply the 4Ps or 7Ps to promote biomass products.

## Summarize

Summarize the topics covered in the module. Participants role-play as business owners and customers.

Create scenarios where they must apply the 4Ps or 7Ps to promote biomass products. You may ask the learners to recall what all has been taught.

## UNIT 3.2: Sales Planning and Customer Engagement

### Unit Objectives

At the end of this unit, you will be able to:

1. Learn how to make an estimate of sale of biogas or biomass pellet to the industry or individual
2. Describe sales process and sales steps.
3. Identify customer satisfaction and service tracking techniques.

### Say

Welcome to Unit 3.2: Sales Planning and Customer Engagement. Today, we're diving into the biomass and bioenergy sector—a growing field focused on sustainable energy solutions. Whether you're selling biogas to a factory or biomass pellets to a household, success depends on solid planning and strong customer relationships. By the end of this session, you'll know how to estimate sales, follow a clear sales process, and keep customers happy. Let's get started!

### Ask

- Have you ever had to convince someone to buy something? What worked or didn't work?
- What do you think industries or households look for when choosing an energy source?
- Why do you think keeping customers satisfied matters in a business-like bioenergy?

### Explain

#### Biogas & Biomass Pellet Sales Estimation

##### Key Points:

- **Market Demand:** Industrial (e.g., cement, textiles) vs. Individual (e.g., households, small businesses).
- **Estimation Methods:** Consumption-based, historical data, competitive analysis, government policies.
- **Pricing & Revenue:** Cost components (raw materials, production, transport) and revenue projection.
- **Financial Viability:** Break-even analysis and ROI.

##### Example:

"Imagine a textile factory using coal. You'd analyze their energy needs, show how biogas cuts costs and emissions, and estimate sales based on their boiler capacity."

#### Sales Process and Sales Steps (20 minutes)

##### Key Points:

- Steps: Identify customers, understand needs, present product, negotiate price, close sale, deliver/install, provide after-sales support.
- Tailor to bioenergy: Highlight sustainability, cost savings, and subsidies.

**Example:**

For a rural household, you'd demo a biogas digester, explain how it uses waste, and offer a payment plan to close the sale.

**Customer Satisfaction & Service Tracking Methods (20 minutes)****Key Points:**

- **Satisfaction Techniques:** Surveys, NPS (explain calculation), reviews, social media, support analysis, focus groups.
- **Service Tracking:** CRM systems, service quality metrics (e.g., response time), loyalty programs, mystery shopping, follow-ups.

**Example:**

"After selling biomass pellets, a quick survey asking 'How likely are you to recommend us?' gives you an NPS score to track satisfaction.

## Notes for Facilitation

- Encourage participants to share personal experiences briefly.
- Write key points on a whiteboard/flipchart to refer to later (e.g., cost, convenience, trust).
- Transition to: "These ideas connect directly to what we'll cover today—estimating demand, selling effectively, and ensuring satisfaction.
- Use props (e.g., a sample pellet bag) if available.
- Provide calculators or worksheets for revenue calculation (e.g.,  $100 \times 5 \times 0.20 \times 30 = \$300/\text{month}$ ).
- Highlight creative survey ideas during presentations.

## Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play,

**Activity:**

- **Setup:** Divide participants into pairs—one as salesperson, one as customer (e.g., factory manager, farmer, homeowner).
- **Task:** Sell biogas or biomass pellets using the sales steps (identify, present, negotiate, close).
- **Scenarios:**
  1. Factory needing cheaper fuel.
  2. Household wanting a cooking solution.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

Today, we we learned how to estimate sales by analyzing demand and pricing, walked through the sales process from identifying customers to after-sales support, and explored ways to track satisfaction like NPS and surveys. These skills are your toolkit for succeeding in the bioenergy sector. Any final questions?"



## UNIT 3.3: Market Research and Performance Assessment

### Unit Objectives

At the end of this unit, you will be able to:

1. Discuss global marketing and the potential for MSMEs as global enterprises.
2. Identify criteria to assess marketing performance.
3. Understand to use primary and secondary research to profile your target customer like thermal power stations, Industries and households.

### Say

Welcome to Unit 3.3: Market Research and Performance Assessment. In today's session, we're exploring how businesses in the biomass and bioenergy sector can grow by understanding markets and measuring success. For small businesses—or MSMEs—this is about tapping into global opportunities, figuring out what works in marketing, and knowing your customers inside out. By the end, you'll have tools to take your bioenergy business to the next level, whether you're targeting power plants or households. Let's dive in!

### Ask

- What do you think makes a small business ready to sell globally?"
- "How would you know if a marketing campaign is successful?"
- "What's one way you'd find out what a customer like a factory or a family wants

### Explain

#### Global Marketing and MSMEs as Global Enterprises (20 minutes)

##### Key Points:

- **Importance:** Biomass/bioenergy's role in global renewable energy trends.
- **Opportunities:** Exports, tech transfer, carbon credits, consultancy, localized solutions.
- **Challenges:** Funding, regulations, competition.
- **Strategies:** Research, branding, partnerships, certifications, training.

##### Example:

"A small biogas plant could export digesters to a country pushing for rural energy solutions, partnering with a local distributor to break in."

#### Criteria to Assess Marketing Performance (20 minutes)

##### Key Points:

- **Why Assess:** Understand trends, measure ROI, improve strategies.
- **Criteria:** Sales growth, market share, customer acquisition/retention, lead conversion, ROI, brand awareness, satisfaction, digital metrics, channel performance, sustainability impact.
- **Methods:** Surveys, analytics, CRM, financial reports.

**Example:**

“If your pellet sales jumped 20% after a trade fair, that’s a sign your marketing worked—track it with sales data and customer feedback.”

**Primary and Secondary Research for Customer Profiling (20 minutes)**

**Key Points:**

- **Why Profile:** Tailor offerings, optimize marketing.
- **Primary:** Surveys, interviews, focus groups, observation, pilot studies.
- **Secondary:** Reports, government data, competitor analysis, journals, databases.
- **Targets:** Power stations (fuel needs), industries (cost savings), households (convenience).

**Example:**

“Survey a power plant about coal use, then check government reports on biomass incentives to build their profile.”

## Notes for Facilitation

- Invite short responses and note key ideas (e.g., “partnerships,” “sales,” “surveys”) on a whiteboard.
- Link answers to the session: “Great thoughts—these connect to global strategies, performance metrics, and research we’ll cover today.”
- Keep it light and inclusive to warm up the group.

Use visuals (e.g., a chart of KPIs or a sample survey) to break up text.

Pause for quick questions after each sub-section (e.g., “What’s one global opportunity you see for bioenergy?”).

Tie examples to learners’ contexts (e.g., local power stations or industries).

Provide a question guide (e.g., “What’s your budget?” “Any green goals?”).

Offer feedback: “Nice pitch—adding a cost-saving stat could seal it!”

Debrief: “What did you learn about your customer?”

## Ask

Ask the participants to watch the video.

### VIDEO

Topic: How To Measure Marketing Performance

By: Square 2

Link: <https://www.youtube.com/watch?v=Xs6FNIIZO1M>

Duration: 03m 39 sec



Scan Qr to watch video

## Role Play

Educate them on how to ask the previously explained pertinent questions.

### Activity:

- **Setup:** Pair participants—one as a researcher/marketer, one as a customer (power station manager, industry owner, or homeowner).
- **Task:** Researcher asks 3-5 questions to profile the customer (e.g., “What’s your energy need?”), then pitches a bioenergy solution and suggests one KPI to track success.
- **Scenarios:**
  1. Power station switching from coal.
  2. Household seeking cheaper cooking fuel.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

- Today, we’ve explored how MSMEs can go global with bioenergy, using strategies like partnerships and research. We’ve identified KPIs like sales and satisfaction to measure marketing success. And we’ve learned how primary and secondary research help profile customers like power stations or households. These tools will help you grow smarter and stronger. Any last thoughts or questions?
- You may ask the learners to recall what all has been taught.

## UNIT 3.4: Organic Farming Economics and Sales Models

### Unit Objectives

At the end of this unit, you will be able to:

1. Calculate the cost of production conventional production system over organic production system of farming
2. Estimate the profit for organic farming by selling the product directly to the consumers.
3. Compare the cost of production which will get reduced over conventional production system.
4. Procedure for selling the organic products to local market/organic hut.

### Say

Welcome to Unit: Organic Farming Economics and Sales Models. Today, we're diving into the financial side of organic farming and how it ties into the biomass and bioenergy sector. Organic farming isn't just about sustainability—it's also a smart business choice. By the end of this session, you'll know how to calculate costs, estimate profits, spot savings compared to conventional farming, and sell directly to consumers through local markets or organic huts. Let's explore how organic farming can grow your wallet and the planet!

### Ask

1. What do you think costs more in farming—chemicals or labor? Why?
2. Have you ever bought something organic? What made you choose it?
3. How might selling directly to customers change a farmer's profits?

### Explain

#### Calculating the Cost of Production (15 minutes)

##### Key Points:

- **Components:** Inputs (seeds, fertilizers), labor, land prep, irrigation, certification, pest management, post-harvest.
- **Steps:** List inputs, assess labor, evaluate land costs, etc.
- **Comparison:** Conventional (high chemical costs) vs. organic (labor-intensive, certification costs).

**Example:**

For a hectare, conventional might spend ₹15,000 on fertilizers, while organic spends ₹7,000 on compost—big difference!

**Estimating Profit for Organic Farming (15 minutes)****Key Points:**

- **Formula:** Profit = Total Revenue - Total Costs.
- **Steps:** Calculate revenue (price × units sold), sum costs (production, marketing, transport), subtract for profit.
- **Factors:** Demand, quality, sales channels.

**Example:**

Selling 5,000 kg of biomass briquettes at ₹10/kg = ₹50,000 revenue. Costs ₹30,000, so profit is ₹20,000 monthly.

**Comparing Cost Reductions in Organic Farming (15 minutes)****Key Points:**

- **Savings:** Less chemicals, better soil health, efficient water use, lower fuel costs, waste recycling.
- **Example Comparison:** Conventional (₹35,000/ha) vs. Organic (₹22,000/ha) = ₹13,000 savings.

**Example:**

No pesticides in organic farming cuts costs—and recycled waste can even become compost or bioenergy!

**Procedure for Selling to Local Markets/Organic Huts (15 minutes)****Key Points:**

- **Steps:** Prep products, research markets, ensure compliance, set up sales channels, promote, track sales.
- **Channels:** Local markets, organic huts, farmgate sales, partnerships.

**Example:**

Label your organic manure, sell it at a farmers' market, and use social media to draw a crowd.

## Notes for Facilitation

- Encourage quick sharing of experiences (e.g., buying organic or farming costs).
- Jot down key ideas (e.g., “chemicals,” “quality,” “no middlemen”) on a whiteboard.
- Transition: “These ideas connect to what we’ll crunch today—costs, profits, and sales strategies.
- Use visuals (e.g., cost comparison chart, profit formula) on slides or handouts.
- Pause for quick Q&A after each sub-section (e.g., “What’s one cost organic farming cuts?”).
- Relate examples to local crops or markets familiar to learners.

## Demonstrate

Show a real-world application of cost calculation and sales.

### Activity:

- **Scenario:** Facilitator calculates costs and profit for an organic vegetable farm selling directly at a local market.
- **Steps:**
  1. **Costs:** Seeds (₹2,000), compost (₹3,000), labor (₹5,000), transport (₹1,000) = ₹11,000 total.
  2. **Revenue:** 500 kg veggies at ₹30/kg = ₹15,000.
  3. **Profit:** ₹15,000 - ₹11,000 = ₹4,000.
  4. **Sales Pitch:** “Fresh, organic veggies at the market—better taste, no chemicals!”
- Compare to conventional (e.g., ₹14,000 costs with chemicals).

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.

Practice cost estimation and selling skills.

### Activity:

- **Setup:** Pair participants—one as farmer, one as buyer (e.g., local market shopper, organic hut owner).
- **Task:** Farmer estimates costs/profit for a product (e.g., organic rice, biomass briquettes), then pitches it to the buyer.

### Scenarios:

1. Selling organic rice to a health-conscious consumer.
2. Selling biomass feedstock to an organic hut.

Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

We’ve learned to calculate costs for organic vs. conventional farming, estimate profits from direct sales, spot cost savings in organic methods, and plan sales to local markets or huts. These skills help you make money and make a difference sustainably. Any final questions?

You may ask the learners to recall what all has been taught.



## 4. Assess Compliance with Applicable Laws, Policies, and Procedures

Unit 4.1 – Regulatory Framework and Compliance for Bio-Energy Businesses

Unit 4.2 – Environmental Regulations and Pollution Control Compliance

Unit 4.3 – Business Registration and Legal Formalities

Unit 4.4 – Government Schemes and Support for Bio-Energy Ventures





## Key Learning Outcomes

**At the end of this module, you will be able to:**

1. Identify the local and central authorities responsible for regulating bio-energy businesses.
2. Explain the role of regulatory bodies like the Pollution Control Board, Municipal Corporations, and Environmental Departments.
3. Understand the key provisions of the Waste Management Rules applicable to bio-energy units.
4. Explain the responsibilities of businesses under the Hazardous Waste Management and Transboundary Movement Rules, 2016.
5. Demonstrate effective methods of coordinating with government authorities and local bodies for regulatory compliance.
6. Apply knowledge of procedures to ensure timely approvals and clearances.
7. Explain the step-by-step procedure to obtain pollution control clearance from the State or Central Pollution Control Board.
8. Evaluate the importance of environmental impact assessments and mitigation measures.
9. Discuss the applicable statutory laws, environmental policies, and industry standards that govern bio-energy operations.
10. Ensure compliance with permits, licenses, and legal obligations for bio-energy businesses.
11. Identify and describe key government schemes supporting bio-energy development in the country.
12. Explain the eligibility criteria, benefits, and application procedures for availing these schemes.
13. Demonstrate the procedures for business registration, including obtaining licenses, GST registration, and MSME certifications.
14. Ensure proper documentation and legal compliance during business setup.
15. Understand the importance of transparent record-keeping for audits and inspections.

## UNIT 4.1: Regulatory Framework and Compliance for Bio-Energy Businesses

### Unit Objectives

At the end of this unit, you will be able to:

1. Identify local authorities having a bearing on business process
2. Discuss applicable statutory laws, policies, standard, permissions and compliances.

### Say

Welcome to Unit: Regulatory Framework and Compliance for Bio-Energy Businesses. The bioenergy sector is key to sustainable energy, but it doesn't operate in a vacuum—rules and regulations shape how businesses run. Today, we'll explore the local authorities you'll work with and the laws, policies, and standards you need to follow. By the end, you'll know who's in charge and what's required to keep your bioenergy venture legal and thriving. Let's get started!

### Ask

- Have you ever dealt with a government office for a project? What was it like?
- What rules do you think a bioenergy plant might need to follow?
- Why might compliance matter for a business using biomass?

### Explain

#### Overview of Local Authorities Impacting Business Operations (30 minutes)

##### Key Points:

- **Authorities:** State Pollution Control Boards (SPCBs), State Electricity Regulatory Commissions (SERCs), State Nodal Agencies (SNAs), Urban Local Bodies (ULBs), Forest/Agricultural Departments, State Industrial Development Corporations (SIDCs), Fire/Safety Departments, Labour Departments, Gram Panchayats.
- **Roles:** Issue permits (e.g., pollution, grid connectivity), provide support (e.g., subsidies, land), enforce safety and labor laws.

##### Example:

"A biomass pellet plant needs SPCB clearance for emissions, SERC approval for selling power, and ULB support for waste supply."

#### 4.1.2 Key Statutory Laws, Policies, and Compliance Requirements (30 minutes)

##### Key Points:

- **Laws:** Environment Protection Act (1986), Air/Water Acts, Electricity Act (2003), Forest Conservation Act (1980), Waste Management Rules.

- **Policies:** National Bio-Energy Mission, National Policy on Biofuels (2018), state renewable energy policies.
- **Standards:** BIS, CPCB emission norms, ISO 14001/50001.
- **Permissions:** Consent to Establish/Operate, grid connectivity, EIA, land use.
- **Compliance:** Emission monitoring, energy audits, tax/labor adherence.

**Example:**

“To start a biogas plant, you’d need an EIA under the Environment Protection Act and a PPA under the Electricity Act.

## Notes for Facilitation

- Encourage short responses and note ideas (e.g., “permits,” “pollution,” “safety”) on a whiteboard.
- Link to the session: “These experiences and guesses tie into the authorities and regulations we’ll cover today.”
- Keep it casual to build rapport.
- Provide a cheat sheet (e.g., “SPCB needs emission data; SERC needs PPA details”).
- Offer feedback: “Good request—next time, mention safety measures too.”
- Debrief: “How did it feel to navigate rules?”

## Demonstrate

Show a real-world application of regulatory navigation.

**Activity:**

- **Scenario:** Facilitator walks through setting up a small biomass power plant.
- **Steps:**
  1. **Authorities:** Contact SPCB for pollution permits, SERC for power sales, and ULB for waste supply.
  2. **Laws/Permits:** Show a mock Consent to Operate form and mention the Air Act.
  3. **Compliance:** Highlight emission monitoring with a fake report (“CO2 levels below limit”).
- End with: “This plant’s now legal and ready to run!”

## Team Activity

- Divide the class into five to six groups of four participants each.
- Pick a bioenergy project (e.g., biomass briquette plant, waste-to-energy unit).
- List 3 authorities to approach and 2 permits needed.
- Suggest 1 compliance step (e.g., emission testing).
- Give the groups 10 minutes for the activity

## Summarize

Summarize the topics covered in the module.

- Today, we identified key local authorities like SPCBs and SERCs that impact bioenergy businesses. We also discussed laws like the Environment Protection Act, policies like the Bio-Energy Mission, and compliance needs like permits and audits. These keep your operations legal and sustainable. Any last questions?”
- You may ask the learners to recall what all has been taught.

## UNIT 4.2: Environmental Regulations and Pollution Control Compliance

### Unit Objectives

At the end of this unit, you will be able to:

1. Explain waste management rules associated with the unit
2. Explain hazardous waste management and trans boundary movement rule 2016
3. Explain procedures to obtain clearance of pollution control board.

### Say

Welcome to Unit: Environmental Regulations and Pollution Control Compliance. In the biomass and bioenergy sector, keeping the environment safe is as important as generating energy. Today, we'll dive into the rules for managing waste, handling hazardous materials, and getting Pollution Control Board approvals. By the end, you'll know how to stay compliant, protect the planet, and keep your operations running smoothly. Let's get started!

### Ask

What kinds of waste do you think a biomass plant might produce?

Why might hazardous waste need special rules?

Have you ever had to get a permit for something? What was the process like?

### Explain

#### Waste Management Rules Associated with the Unit (20 minutes)

##### Key Points:

- **Waste Types:** Agricultural residues, processing waste (e.g., bagasse), combustion by-products (e.g., ash), effluents, hazardous waste (e.g., oils).
- **Rules:** Solid Waste Management Rules (2016), Hazardous Waste Rules (2016), Bio-Medical/E-Waste Rules (2016), Water/Air Acts (1974/1981), Environment Protection Act (1986).
- **Steps:** Audit, classify, get permits, store/treat safely, keep records.
- **Best Practices:** Reuse ash, compost organics, recycle.

##### Example:

A biogas plant might compost leftover sludge and send ash to a brick factory—smart and compliant!

### Hazardous Waste Management and Transboundary Movement Rules, 2016 (20 minutes)

#### Key Points:

- **Hazardous Waste:** Ash/sludge, used oils, chemical residues, e-waste.
- **Rules:** Classify, get SPCB authorization, store/label properly, transport safely, keep records, report accidents.
- **Transboundary:** Prior Informed Consent, environmentally sound disposal, import restrictions.
- **Compliance:** Audits, training, authorized processors.

#### Example:

If a biomass plant generates used oil, it needs labeled drums and an SPCB permit—no dumping allowed!

### Procedures to Obtain Clearance of Pollution Control Board (20 minutes)

#### Key Points:

- **Steps:** Check category (e.g., Orange), apply for Consent to Establish (CTE) and Operate (CTO), submit docs/EIA, get NOC, install controls, final inspection, monitor compliance.
- **Documents:** Project report, site plan, waste plans, pollution controls, NOC from local bodies.

#### Example:

For a pellet plant, you'd submit a site plan to the SPCB, install a bag filter, and get CTO after inspection.

## Notes for Facilitation

- Encourage quick answers and note ideas (e.g., “ash,” “safety,” “paperwork”) on a whiteboard.
- Tie responses to the session: “These connect to the waste rules and PCB clearances we’ll explore today.”
- Keep it light to build engagement.
- Provide a reference sheet (e.g., waste rules, CTE/CTO steps).
- Encourage detail (e.g., “Ash under Solid Waste Rules, CTE needs EIA”).
- Highlight thorough or creative plans during shares.

## Role Play

#### Activity:

- **Setup:** Pair participants—one as a bioenergy unit manager, one as an SPCB officer or waste handler.
- **Task:** Manager explains waste plans (e.g., “We compost sludge, store oil safely”) and requests CTE/CTO, officer responds with requirements.
- **Scenarios:**

1. Biomass briquette plant seeking CTE.
  2. Biogas unit arranging hazardous waste pickup.
- Switch roles after 10 minutes.

## Demonstrate

Show video

### VIDEO

Topic: CTE / CTO Consent to Established  
/ Operate, CPCC / CPCB, SPCB, Pollution index  
By: HUB of KNOWLEDGE MECH

Link: <https://www.youtube.com/watch?v=3Wh5pwNHt7E>

Duration: 12 m 32 sec



Scan Qr to watch video

## Summarize

Summarize the topics covered in the module.

- Today, we've covered waste management rules like Solid Waste and Hazardous Waste Rules, dug into the 2016 Hazardous Waste regulations—including transboundary movement—and walked through getting PCB clearance with CTE and CTO. These keep your bioenergy unit legal and eco-friendly. Any final questions?
- You may ask the learners to recall what all has been taught.

## UNIT 4.3: Business Registration and Legal Formalities

### Unit Objectives

At the end of this unit, you will be able to:

1. Explain how to co-ordinate with government authorities/local authorities, corporations etc. to comply with their regulations
2. Show them how to maintain records.
3. Demonstrate methods of completing business registration, associated legal formalities and compliances.

### Say

Welcome to Unit: Environmental Regulations and Pollution Control Compliance. In the biomass and bioenergy sector, keeping the environment safe is as important as generating energy. Today, we'll dive into the rules for managing waste, handling hazardous materials, and getting Pollution Control Board approvals. By the end, you'll know how to stay compliant, protect the planet, and keep your operations running smoothly. Let's get started!

### Ask

Have you ever had to deal with a government office for a permit or approval? How did it go?

Why do you think keeping records matters for a business?

What steps do you imagine are needed to register a company?

### Explain

#### Coordinating with Government Authorities, Local Bodies, and Corporations

##### Key Points:

- **Authorities:** MNRE, SREAs, SPCBs, municipal corporations, panchayats, corporations.
- **Steps:** Identify authorities, get approvals (e.g., environmental clearance, RECs), communicate effectively, document everything, build relationships, monitor compliance.

##### Example:

"To set up a biogas plant, you'd get a CTE from the SPCB and partner with a local corporation for waste supply."

#### Maintaining Records



#### Key Points:

- **Why:** Compliance, audits, transparency, decision-making.
- **Types:** Registration (e.g., GST certificate), legal (e.g., CTE/CTO), financial (e.g., tax returns), operational (e.g., production logs), HR (e.g., payroll).
- **Best Practices:** Organize, update, secure, check compliance, limit access.
- **Tools:** Tally, Google Drive, ComplianceQuest.

#### Example:

“Store your SPCB clearance digitally and keep a physical copy—ready for an audit anytime!”

#### Methods of Completing Business Registration, Legal Formalities, and Compliances

#### Key Points:

- **Registration:** Choose structure (sole prop, partnership, Pvt Ltd, LLP), get name approval, obtain PAN/TAN/GST/factory license.
- **Formalities:** Environmental clearances (CTE/CTO, EIA), land permits, safety/labor compliance (ESIC, EPFO).
- **Compliances:** RPO, waste rules (Solid/Hazardous Waste), financial filings (GST, MCA returns).

#### Example:

“For a biomass briquette unit, register as a Pvt Ltd via SPICe+, get a GST number, and secure a CTE from the SPCB.

## Notes for Facilitation

- Invite quick responses and jot down ideas (e.g., “permits,” “proof,” “forms”) on a whiteboard.
- Link to the session: These experiences tie into coordinating with authorities, record-keeping, and registration we’ll cover today.
- Keep it relaxed to warm up the group.
- Use visuals (e.g., authority list, record types table, registration flowchart) on slides or handouts.
- Pause for quick Q&A (e.g., “Which authority handles land permits?”).

## Team Activity

- Divide the class into five to six groups of four participants each.

Pick a bioenergy business (e.g., biofuel unit, pellet plant).

List 2 authorities to coordinate with and 1 approval each needs.

Name 3 records to maintain and 2 registration steps.

**Output:** Groups present their plan in 10 minutes.

## Demonstrate

Show a video

### VIDEO

Topic: How to Register a Company in India:  
Business Structure, Documentation,  
Legal Compliance and Process

By: Backstage with Millionaires

Link: <https://www.youtube.com/watch?v=KjsFM0lxQYk>

Duration: 8 m 44 sec



Scan Qr to watch video

## Summarize

Summarize the topics covered in the module.

- Today, We've learned how to coordinate with authorities like SPCBs and municipalities, maintain records like permits and financials, and register a business with steps like SPICe+ and CTE/CTO. These keep your bioenergy venture legal and credible. Any last questions?
- You may ask the learners to recall what all has been taught.

## UNIT 4.4: Government Schemes and Support for Bio-Energy Ventures

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain different schemes supporting Bio-energy in the country.
2. Illustrate key highlights of schemes supporting bioenergy in the country.

### Say

Welcome to Unit: Government Schemes and Support for Bio-Energy Ventures. The government is a big ally in growing bio-energy businesses, offering schemes that provide money, tech help, and more. Today, we'll explore the key programs supporting biomass and bio-energy in India and highlight what makes them valuable. By the end, you'll know how to tap into these opportunities to kickstart or scale your venture. Let's get going!

### Ask

- Have you heard of any government programs that help renewable energy projects? Which ones?
- What kind of support do you think a bio-energy business might need from the government?
- Why might government backing matter for something like biogas or biomass?

### Explain

#### **Different Schemes Supporting Bio-Energy in the Country**

##### **Key Points:**

- **Schemes:**
  - National Bio-Energy Mission (NBEM): Biomass power, waste-to-energy, R&D.
  - Biomass Power & Cogeneration Program (BPCP): Bagasse cogeneration, CFA.
  - Waste to Energy Program: Biogas, biomethanation, subsidies.
  - National Bio-Energy Fund: Loans, grants.
  - SATAT: CBG production, purchase agreements.
  - GOBAR-Dhan: Rural biogas, organic fertilizers.
  - RETD: R&D funding.

- State Initiatives: Subsidies, tax breaks (e.g., Maharashtra, Punjab).
- **Support:** Financial aid, technical assistance, policy backing.
- Example:**  
“SATAT helps set up CBG plants with guaranteed buyers—like a safety net for biogas entrepreneurs!”
- Key Highlights of Schemes Supporting Biomass/Bio-Energy**
- Key Points:**
- **Highlights:**
  - NBEM: Broad support, R&D focus.
  - BPCP: Bagasse power, rural jobs.
  - Waste to Energy: Waste utilization, tech support.
  - National Bio-Energy Fund: Low-cost funding.
  - SATAT: Bio-CNG push, emission cuts.
  - GOBAR-Dhan: Rural wealth, sustainability.
  - RETD: Innovation boost.
  - State Initiatives: Localized perks.
- **Impact:** Sustainability, jobs, clean energy growth.
- Example:**  
“GOBAR-Dhan turns cow dung into biogas and fertilizer—rural income plus green energy!”

## Notes for Facilitation

- Encourage quick sharing and note ideas (e.g., “subsidies,” “tech,” “sustainability”) on a whiteboard.
- Link to the session: “These ideas connect to the schemes and benefits we’ll cover today.
- Keep it casual to build engagement.
- Provide a guide (e.g., “SATAT offers purchase agreements; NBEM gives CFA”).
- Offer feedback: “Great pitch—mention waste volume next time!”
- Debrief: “How did it feel to pitch for support?”

## Team Activity

- Divide the class into five to six groups of four participants each.

### Task:

Pick a bio-energy project (e.g., biogas, biomass power).  
Choose 2 schemes to apply for and list 1 benefit each.  
Outline 2 steps to access the schemes.  
Give the groups 5 minutes for the activity

## Demonstrate

- Show a video

### VIDEO

Topic: Complete Government Schemes Course –

GOBARdhan scheme, SATAT Scheme,

National Bioenergy Programme

By: Govt Jobs Adda247

Link: <https://www.youtube.com/watch?v=cFtU0xWjqyw>

Duration: 26 m 48 sec



Scan Qr to watch video

## Summarize

Summarize the topics covered in the module.

- Today, We've explored schemes like NBEM, SATAT, and GOBAR-Dhan that support bio-energy with funding, tech, and more. We've highlighted their perks—like rural jobs and clean fuel. These tools can power your bio-energy dreams. Any final questions?
- You may ask the learners to recall what all has been taught.





## Key Learning Outcomes

**At the end of this module, you will be able to:**

1. Identify the requirements for a safe work area in a bioenergy manufacturing facility, including hazard-free zones, emergency exits, fire safety measures, and proper ventilation.
2. Explain occupational health & safety standards and regulations specific to bioenergy manufacturing to ensure compliance with legal and workplace guidelines.
3. Recognize work safety procedures and instructions for operating bioenergy machinery safely and efficiently, including biomass processing, gasification, and biogas generation units.
4. Explain how to administer first aid in case of workplace injuries such as burns, chemical exposure, or electric shocks.
5. Demonstrate first aid techniques and the correct use of first aid kits in a bioenergy manufacturing setup.
6. Identify emergency protocols to respond to workplace incidents like fires, gas leaks, and equipment malfunctions promptly and effectively.
7. Identify different types of personal protective equipment (PPE) used in bioenergy facilities, including flame-resistant clothing, gloves, goggles, respirators, and safety shoes.
8. Illustrate the proper use of PPE for various tasks in bioenergy operations to prevent injuries and health hazards.
9. Demonstrate how to identify and follow work safety procedures for operating bioenergy machinery while ensuring operational efficiency.
10. Implement lockout/tagout (LOTOTO) procedures to prevent accidental machine start-up during maintenance and repair activities.
11. Demonstrate good housekeeping practices to maintain a clean and organized work environment in bioenergy plants, ensuring safety and operational efficiency.
12. Implement infection control guidelines and biohazard management protocols to minimize contamination and health risks in the workplace.
13. Demonstrate how to manage and dispose of waste materials like biomass residue, ash, and other by-products according to safety and environmental regulations.



## UNIT 5.1: Workplace Safety and Hazard Identification in Bioenergy Manufacturing

### Unit Objectives

At the end of this unit, you will be able to:

1. Identify the requirements for a safe work area in a bioenergy manufacturing facility, including hazard-free zones, emergency exits, fire safety measures, and proper ventilation.
2. Explain occupational health & safety standards and regulations specific to bioenergy manufacturing to ensure compliance with legal and workplace guidelines.
3. Recognize work safety procedures and instructions for operating bioenergy machinery safely and efficiently, including biomass processing, gasification, and biogas generation units.

### Say

Welcome to Unit: Workplace Safety and Hazard Identification in Bioenergy Manufacturing. Safety is the backbone of any bioenergy facility—whether you’re processing biomass, running gasifiers, or generating biogas, the risks are real. Today, we’ll learn how to create a safe work area, understand key safety regulations, and operate machinery without putting ourselves or others in danger. By the end, you’ll have the tools to keep a bioenergy plant running smoothly and safely. Let’s get started!

### Ask

- What’s one safety risk you’ve seen or heard about in a workplace?
- Why do you think safety rules matter in a place with machines and gases?
- What would you do if you smelled gas in a bioenergy plant?

### Explain

#### Requirements for a Safe Work Area

##### Key Points:

- **Hazard-Free Zones:** Clear signage, no obstructions, safe storage.
- **Emergency Exits:** Well-lit, accessible, mapped routes.
- **Fire Safety:** Extinguishers, sprinklers, flammable storage rules.
- **Ventilation:** Exhaust systems, air quality monitoring.

##### Example:

“Imagine a biomass storage area—piles of wood chips could spark a fire. Keeping it ventilated and away from machines, with extinguishers nearby, makes it a safe zone.”

#### Occupational Health & Safety Standards and Regulations

##### Key Points:

- **Importance:** Protects workers, environment, ensures compliance.
- **Hazards:** Chemical (gases), fire, physical (machinery), biological, ergonomic.

- **Regulations:** Factories Act, OSHA, EPA, ILO, NFPA, PCB guidelines.
  - **Workplace Guidelines:** Risk assessments, PPE, training, emergency plans.
- Example:**  
“OSHA says you need PPE like respirators near biogas tanks to avoid methane exposure—non-negotiable for safety and the law.”
- Work Safety Procedures for Bioenergy Machinery**
- Key Points:**
- **General Guidelines:** Training, PPE, signage, housekeeping.
  - **Biomass Processing:** Pre-checks, no loose clothing, dust control.
  - **Gasification:** Monitor pressure, avoid gas exposure, LOTO.
  - **Biogas Units:** Gas detectors, no hot work, monitor digestion.
  - **Maintenance:** Routine checks, certified repairs.
- Example:**  
“Before starting a gasifier, check the pressure valves. One leak could mean carbon monoxide trouble—safety first!”

## Notes for Facilitation

- Invite brief responses and note ideas (e.g., “fire,” “training,” “evacuate”) on a whiteboard.
- Link to the session: “These risks—like gas leaks or machine hazards—are exactly what we’ll tackle today.”
- Keep it conversational to build engagement.
- Provide a checklist (e.g., “Check valves, wear PPE, monitor gauges”).
- Offer feedback: “Good shutdown—next time, call out the hazard louder!”
- Debrief: “What felt most critical to safety?”

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to complete a mock packing and dispatch process, including preparing labels, verifying documents, and coordinating logistics.
- Ask the group to discuss.

Give the groups 5 minutes for the activity.

## Role Play

Educate them on how to ask the previously explained pertinent questions.

**Setup:** Pair participants—one as operator, one as supervisor.

**Task:** Operator performs a safety check and runs a machine (e.g., gasifier or biogas unit), while supervisor ensures compliance and responds to a mock hazard (e.g., gas leak).

**Scenarios:**

1. Gasifier overheating—shut it down safely.
  2. Biogas leak—evacuate and report.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

We've identified how to make a bioenergy work area safe with zones, exits, fire measures, and ventilation. We've explored OHS standards like OSHA to stay legal and safe. And we've learned procedures to run machinery without risks. These skills keep you and your team safe and productive. Any last thoughts?

You may ask the learners to recall what all has been taught.

## UNIT 5.2: First Aid and Emergency Response in Bioenergy manufacturing Facilities

### Unit Objectives

At the end of this unit, you will be able to:

1. Explain how to administer first aid in case of workplace injuries such as cuts, burns, or electric shocks.
2. Identify emergency protocols to respond to workplace incidents promptly and effectively.

### Say

Welcome to Unit: First Aid and Emergency Response in Bioenergy Manufacturing Facilities. In a bioenergy plant, where we deal with machinery, heat, and hazardous materials, accidents can happen. Knowing how to respond—whether it's treating a cut or evacuating during a fire—can save lives and keep the workplace safe. Today, you'll learn to administer first aid for common injuries and follow emergency protocols to act fast and smart. Let's prepare to be the calm in the storm!

### Ask

- Have you ever had to help someone who was hurt? What did you do?
- What's the first thing you'd do if you saw a coworker get burned?
- Why do you think quick action matters in an emergency?

### Explain

#### Administering First Aid for Workplace Injuries

##### Key Points:

- **Cuts/Wounds:** Assess severity, clean, apply pressure, bandage, seek help if deep.
- **Burns:** Remove from source, cool with water (10-15 min), cover, no ice/butter, medical help for severe cases.
- **Electric Shock:** Ensure safety, cut power, move with non-conductive object, check breathing (CPR if needed), treat burns, get help.

##### Example:

"If a worker cuts their hand on a shredder, you'd rinse it, press with gauze to stop bleeding, and bandage it—quick and clean keeps it from getting worse."

#### Emergency Protocols to Respond to Workplace Incidents

##### Key Points:

- **Reporting:** Notify supervisor, use alarms, give clear details.
- **Evacuation:** Follow routes, keep exits clear, use wardens.
- **Fire Safety:** Sound alarm, use extinguishers, evacuate if uncontrolled.

- **First Aid/Medical:** Treat on-site, keep kits stocked, call pros for severe cases.
- **Hazmat:** Use MSDS, PPE, isolate spills.
- **Post-Emergency:** Investigate, record, support workers, improve.

**Example:**

“A gas leak triggers an alarm—you report it, evacuate via marked exits, and let the hazmat team handle it while you grab a first aid kit for anyone affected.

## Notes for Facilitation

- Encourage brief stories or guesses (e.g., “I’d run for help”).
- Note key ideas (e.g., “stop bleeding,” “call for help”) on a whiteboard.
- Transition: “These instincts are spot on—today, we’ll build on them with specific skills and plans.

## Demonstrate

Show a video

### VIDEO

Topic: Administering First Aid

By: dchummer Chemistry (2009)

Link: <https://www.youtube.com/watch?v=7TSqfYfdlb8>

Duration: 4 m 20 sec



Scan Qr to watch video

## Role Play

### Activity:

- **Setup:** Pair participants—one as responder, one as victim.
  - **Task:** Responder treats an injury (cut, burn, or shock) and initiates an emergency protocol (e.g., report, evacuate).
  - **Scenarios:**
    1. Cut from a biomass shredder—clean and bandage, report to supervisor.
    2. Burn from a steam leak—cool and cover, sound fire alarm.
    3. Shock from a panel—cut power, check breathing, call for help.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

- Today, We've learned to administer first aid for cuts, burns, and shocks—cleaning, cooling, and checking vitals can make all the difference. We've also set up emergency protocols to report, evacuate, and respond fast to incidents. These skills keep a bioenergy facility safe and ready. Any final questions?
- You may ask the learners to recall what all has been taught.

## UNIT 5.3: Safety Practices and Equipment Protection in Bioenergy Operations

### Unit Objectives

At the end of this unit, you will be able to:

1. Identify different types of personal protective equipment (PPE) used in Bioenergy manufacturing facility, including gloves, goggles, respirators, and safety shoes.
2. Illustrate the proper use of PPE for various tasks to prevent injuries and health hazards.
3. Identify heavy electrical and mechanical equipment associated with Bioenergy manufacturing Facility and understand their functions.

### Say

Welcome to Unit 5.3: Safety Practices and Equipment Protection in Bioenergy Operations. In a bioenergy facility, safety isn't optional—it's essential. From handling biomass to running heavy machinery, risks are everywhere. Today, we'll explore the personal protective equipment (PPE) that keeps you safe, how to use it right, and the key equipment that powers these plants. By the end, you'll know how to protect yourself and keep operations running smoothly. Let's gear up and dive in!

### Ask

What's one piece of safety gear you've seen or used at work? How did it help?  
"What might happen if you don't wear the right protection near a machine?"  
"Can you name a machine you'd find in a bioenergy plant?"

### Explain

#### Personal Protective Equipment (PPE) Used in Bioenergy Facilities

##### Key Points:

- **Types:**
  - Gloves: Chemical-resistant, cut-resistant, heat-resistant, insulated.
  - Eye/Face: Goggles, face shields, welding helmets.
  - Respirators: Dust masks, full-face, SCBA.
  - Safety Shoes: Steel-toe, chemical-resistant, slip-resistant.
- **Importance:** Prevents injuries, ensures compliance, boosts confidence.

##### Example:

"Handling biomass? Cut-resistant gloves stop splinters. Near a gasifier? Heat-resistant gloves protect from burns."

#### Proper Use of PPE for Various Tasks

##### Key Points:

- **Gloves:** Secure fit, check for damage (e.g., chemical handling: nitrile gloves, sealed cuffs).

- **Goggles/Face Shields:** Snug fit, clear lenses (e.g., grinding: face shield over goggles).
- **Respirators:** Proper seal, fit test (e.g., confined spaces: SCBA with buddy system).
- **Safety Shoes:** Well-fitted, maintained (e.g., machinery: steel-toe, tied laces).

**Example:**

“Mixing chemicals? Wear nitrile gloves and goggles—tight fit, no gaps—to avoid splashes.”

**Heavy Electrical and Mechanical Equipment**

**Key Points:**

- **Electrical:** Generators (power), transformers (voltage), motors (motion), switchgear (control), VFDs (speed).
- **Mechanical:** Conveyors/chippers (feedstock), boilers (heat), gasifiers (syngas), digesters (biogas), turbines (power), heat exchangers (efficiency), pumps (fluids), flue gas systems (emissions).
- **Importance:** Efficiency, safety, compliance.

**Example:**

“A gasifier turns biomass into syngas for power—knowing it helps you spot risks like gas leaks.

## Notes for Facilitation

Encourage quick responses (e.g., “Gloves stopped a cut,” “A shredder”).

Note ideas (e.g., “goggles,” “injury,” “boiler”) on a whiteboard.

Transition: “These are great starting points—today, we’ll cover PPE and equipment in detail.

Provide a PPE checklist (e.g., “Gloves: snug, no holes”).

Offer feedback: “Good respirator fit—mention gas risks next time!”

Debrief: “What was trickiest about PPE?”

## Demonstrate

Show a video

**VIDEO**

Topic: Personal Protection Equipment

By: National Safety Compliance: OSHA Safety Training (2017)

Link: <https://www.youtube.com/watch?v=YE-63aeECVs>

Duration: 24 m 09 sec



Scan Qr to watch video



## Summarize

Summarize the topics covered in the module.

- Today, we've identified PPE like gloves and respirators for bioenergy work, learned how to use them right for tasks like chemical handling, and explored equipment like gasifiers and turbines that power these plants. These skills keep you safe and the facility humming. Any last questions?
- You may ask the learners to recall what all has been taught.

## UNIT 5.4: Waste Disposal and Housekeeping Practices

### Unit Objectives

At the end of this unit, you will be able to:

1. Demonstrate good housekeeping practices to maintain a clean and organized work environment.
2. Implement infection control guidelines to minimize contamination and health risks in the workplace.
3. Demonstrate how to dispose of waste materials according to safety and environmental regulations.

### Say

Welcome to Unit: Waste Disposal and Housekeeping Practices. In a bioenergy facility, keeping things clean and managing waste isn't just about tidiness—it's about safety, health, and staying legal. From biomass scraps to chemical spills, how we handle waste and maintain our workspace can prevent accidents and protect the environment. Today, you'll learn housekeeping tricks, infection control basics, and how to dispose of waste the right way. Let's clean up and stay safe!

### Ask

- What's one messy thing you've seen at a workplace that could cause trouble?
- How do you think germs or contamination could spread in a bioenergy plant?
- What's one type of waste you'd expect from making bioenergy.

### Explain

Tell the learner

#### Housekeeping Practices

##### Key Points:

- **Importance:** Safety, efficiency, compliance.
- **5S Methodology:** Sort, Set, Shine, Standardize, Sustain.
- **Practices:** Clean floors/machines, manage waste, organize tools, clear pathways, store chemicals safely.

##### Example:

"A cluttered walkway near a gasifier could trip someone—keeping it clear with 5S prevents that."

#### Infection Control Guidelines

##### Key Points:

- **Risks:** Biological (waste), chemical (solvents), airborne (dust), waterborne (pathogens).

- **Guidelines:** Hand hygiene, PPE (gloves, masks), disinfect surfaces, manage waste, ventilate, monitor health.
- **Emergency:** Isolate, treat, report contamination.

**Example:**

“Biomass residue can grow mold—washing hands and wearing a mask stops you from breathing it in.”

**Waste Disposal Practices**

**Key Points:**

- **Types:** Organic (residues), chemical (solvents), solid (packaging), hazardous (toxics), wastewater.
- **Steps:** Classify, segregate, store, transport, treat/dispose, prepare for spills, document.
- **Regulations:** PCB, environmental standards.

**Example:**

“Ash from a boiler goes in a sealed bin, not the trash—then off to a landfill per regs.”

## Notes for Facilitation

- Invite quick answers (e.g., “Spilled oil,” “Dirty tools,” “Ash”).
- Note ideas (e.g., “clutter,” “dust,” “chemicals”) on a whiteboard.
- Transition: “These are real issues we’ll tackle today with practical solutions.”
- Use visuals (e.g., 5S chart, waste bins, PPE) on slides or handouts.
- Pause for quick Q&A (e.g., “What’s one 5S step?”).
- Relate to bioenergy (e.g., local waste like wood chips)

## Demonstrate

**Activity:**

- **Scenario:** Facilitator demonstrates a housekeeping routine and waste segregation.
- **Steps:**
  1. **Housekeeping:** “Wipe down a mock machine, store tools on a shadow board—clean and tidy.”
  2. **Waste:** “Sort biomass scraps into an organic bin, chemicals into a sealed hazmat container.”
  3. **PPE:** Wear gloves and mask—“This keeps me safe from dust or spills.”

## Team Activity

- Divide the class into five to six groups of four participants each.

**Task:**

Create a housekeeping checklist for a bioenergy area (e.g., gasifier zone).

List 2 infection control steps for that area.

Plan waste disposal for 2 waste types (e.g., ash, wastewater).

Output: Groups present their plan in 10 minutes.

## Summarize

Summarize the topics covered in the module.

- Today, We've learned to keep a bioenergy facility clean and organized with housekeeping like 5S, control infections with hygiene and PPE, and dispose of waste safely per regulations. These practices keep us healthy, safe, and compliant. Any final thoughts?
- You may ask the learners to recall what all has been taught.



## 7. Ensure Manufacturing of Biomass pellet

- Unit 7.1 – Biomass Preparation and Placement for Pellet Manufacturing
- Unit 7.2 – Operations with Machinery and Equipment in Biomass Pellet Production
- Unit 7.3 – Quality Control and Testing of Biomass Pellets
- Unit 7.4 – Post-Production Maintenance and Best Practices in Bioenergy Plants



## Key Learning Outcomes

**At the end of this module, you will be able to:**

1. Understanding the process of placing biomass on dies
2. Importance of biomass quality and preparation techniques
3. Demonstrating proper biomass placement on pellet mill dies
4. Fundamentals of the extrusion process in pellet manufacturing
5. Role of die, speed, temperature, and moisture control
6. Adjusting machine parameters for optimal pellet production
7. Overview of key machines: Dryer, hammer mill, pellet mill, bomb calorimeter, etc.
8. Setting control parameters for efficient operations
9. Demonstrating machine operation and adjustment techniques
10. Cooling process of biomass pellets and backpressure management
11. Standard protocols for pellet testing (Gross Calorific Value, Ash, Moisture)
12. Use of moisture meters and other testing equipment
13. Do's and Don'ts in biomass pellet manufacturing
14. Cleaning and maintenance of machines after production

## UNIT 7.1: Biomass Preparation and Placement for Pellet Manufacturing

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain and demonstrate the correct method of placing biomass on dyes for uniform pellet formation.
2. Describe the extrusion process and its role in pellet manufacturing.

### Say

Welcome to Unit: Biomass Preparation and Placement for Pellet Manufacturing. Turning biomass into pellets is a game-changer for bioenergy—it's renewable, efficient, and sustainable. But to get high-quality pellets, how we prepare and place biomass on the dye, and how we handle the extrusion process, makes all the difference. Today, you'll learn the steps to do it right, see it in action, and understand why it matters. Let's shape some pellets and power the future!

### Ask

- What do you think biomass pellets are used for?"
- "Why might uneven pellets be a problem in manufacturing?"
- "Have you seen or worked with a machine that presses materials? How did it work?"

### Explain

#### **Method of Placing Biomass on Dyes (30 minutes)**

##### **Key Points:**

- **What's a Dye:** Perforated plate (3-12mm holes) shapes pellets with rollers.
- **Importance:** Uniform size, no clogs, efficiency, quality.
- **Steps:**
  1. Prep biomass (crush <5mm, 10-15% moisture).
  2. Inspect dye (clean, right size).
  3. Feed evenly (conveyor, no overload).
  4. Ensure distribution (monitor rollers).
  5. Adjust as needed (feed rate, pressure).

##### **Example:**

"Too much sawdust at once clogs a 6mm dye—feed it steady, and you get perfect fuel pellets."

#### **Extrusion Process and Its Role**

##### **Key Points:**



- **What's Extrusion:** Forcing biomass through dye under pressure/temperature.
  - **Components:** Pellet mill, dye, rollers, feeder, cutter.
  - **Steps:** Prep (grind, moisture), feed, compress/extrude, cut/cool.
  - **Role:** Shapes pellets, boosts density/durability, controls moisture, cuts waste.
- Example:**  
 "Rollers squeeze wood chips through the dye—heat binds it into dense pellets that burn longer.

## Demonstrate

Show video

### VIDEO

Topic: Pellet Cooling Process

By: Clara Wang

Link: [https://youtube.com/shorts/vcyB\\_hLkqwQ?si=JaMaI7zDLLLL](https://youtube.com/shorts/vcyB_hLkqwQ?si=JaMaI7zDLLLL)

Duration: 00 m 31 sec



Scan Qr to watch video

### VIDEO

Topic: The Complete Pelletizing Process of Biomass/Wood Pellet M

By: Kingwood Biomass Wood Pellet Mill

Link: <https://www.youtube.com/watch?v=ON6pl4pjpV0>

Duration: 03 m 12 sec



Scan Qr to watch video

## Notes for Facilitation

- Encourage quick answers (e.g., "Fuel," "Burn poorly," "Like a press").
- Note ideas (e.g., "energy," "consistency," "pressure") on a whiteboard.
- Transition: "These ideas tie right into how we prep biomass and extrude it—let's dive deeper.
- Provide a checklist (e.g., "Crush <5mm, feed steady, monitor pellets").
- Offer feedback: "Good even feed—mention roller pressure next time!"
- Debrief: "What was toughest to get right?"

## Team Activity



- Divide the class into five to six groups of four participants each.

Task:

Plan biomass prep for a pellet type (e.g., fuel, feed).

List 3 steps for dye placement.

Describe 2 extrusion roles (e.g., shape, density).

Output: Groups present their plan in 15 minutes.

## Summarize



Summarize the topics covered in the module.

- Today, We've learned how to place biomass on dyes—prep it, feed it evenly, and monitor for uniform pellets. We've also covered extrusion—how it shapes and strengthens pellets for bioenergy. These skills ensure quality and efficiency. Any final questions?
- You may ask the learners to recall what all has been taught.

## UNIT 7.2: Operations with Machinery and Equipment in Biomass Pellet Production

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Discuss and demonstrate the operations of various machines in plant such as dryer, hammer mill, pellet mill, bomb calorimeter, etc.
2. Explain about speed, temperature and moisture control in pelleting process and show how to Set the control parameters during manufacturing
3. Explain the process of cooling of biomass pellet and Express what is backpressure.
4. Show how to Adjust pellet producing machineries at dye of 3mm, specific speed, temperature and moisture control.

### Say

- Welcome to Unit: Operations with Machinery and Equipment in Biomass Pellet Production. Making biomass pellets is all about precision—using machines like dryers, hammer mills, and pellet mills to turn raw materials into energy-packed pellets. Today, we'll master how these machines work, control key factors like speed and moisture, cool pellets effectively, and adjust a pellet mill for 3mm dyes. By the end, you'll be ready to run a smooth, safe, and efficient production line. Let's get those machines humming!

### Ask

Ask the participants

- What's one machine you think is key to making biomass pellets? Why?
- Why might temperature or moisture mess up pellet quality?
- Have you ever cooled something hot—like food or metal? How did you do it?

### Notes for Facilitation

- Encourage quick responses (e.g., "Dryer," "Too wet clogs it," "Fan").
- Note ideas (e.g., "grinding," "heat," "air") on a whiteboard.
- Transition: "These guesses are spot on—let's see how they fit into pellet production.
- Use visuals (e.g., machine diagrams, parameter chart) on slides or handouts.
- Pause for quick Q&A (e.g., "Why cool pellets?").
- Relate to local biomass (e.g., sawdust, straw).

## Explain

### Operations of Various Machines

#### Key Points:

- **Dryer:** Reduces moisture (load, dry with hot air, discharge).
- **Hammer Mill:** Grinds biomass (feed, grind, screen, collect).
- **Pellet Mill:** Forms pellets (feed, compress, cut, cool).
- **Bomb Calorimeter:** Measures energy (prepare, combust, measure).

#### Example:

“A hammer mill smashes wood chips into dust—small enough for the pellet mill to shape.”

### Speed, Temperature, and Moisture Control

#### Key Points:

- **Speed:** Feeder (20-30 RPM), die (150-200 RPM), rollers (1-3 m/s).
- **Temperature:** Pre-condition (80-100°C), die (90-120°C), cool (40-50°C).
- **Moisture:** Raw (10-15%), post-conditioning (15-17%), final (8-12%).
- **Setting Parameters:** Assess material, adjust speeds, monitor temps/moisture.

#### Example:

“Too fast a die speed? Pellets crumble. Keep it at 200 RPM for balance.”

### Cooling Process and Backpressure

#### Key Points:

- **Cooling:** Load hot pellets (70-90°C), use counterflow air, reduce to 40-50°C, 8-12% moisture.
- **Backpressure:** Air resistance from pellet bed height, speed, or density—manage with fan speed, bed height.

#### Example:

“High backpressure from a thick pellet pile slows cooling—adjust the bed to keep air flowing.”

### Adjusting Machinery for 3mm Die

#### Key Points:

- **Die:** Install 3mm, align rollers (0.1-0.3mm gap).
- **Speed:** Set 250-300 RPM.
- **Temperature:** Die at 80-90°C, use steam if needed.
- **Moisture:** Pre-condition to 10-15%, monitor with sensors.

#### Example:

“For 3mm pellets, a tight roller gap and steady 90°C die temp make small, strong pellets.”

## Role Play

### Activity:

- **Setup:** Pair participants—one as operator, one as supervisor.
- **Task:** Operator runs a mock machine (e.g., dryer, pellet mill), sets parameters (e.g., 3mm dye, 90°C), and explains cooling; supervisor checks settings and safety.
- **Scenarios:**
  1. Drying wet straw—adjust temp to 100°C.
  2. Pelleting with 3mm dye—set 250 RPM, 12% moisture.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

- Today, we've explored how to operate dryers, hammer mills, pellet mills, and calorimeters. We've set speed, temp, and moisture for quality pellets, learned cooling and backpressure, and adjusted a 3mm dye mill. These skills keep production top-notch. Any final questions?
- You may ask the learners to recall what all has been taught.

## UNIT 7.3: Quality Control and Testing of Biomass Pellets

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Explain about standard protocol for testing of pellets and important parameters are Gross Calorific Value, Ash and Moisture.
2. Demonstrate use of moisture meter to measure water content in different agriculture waste.

### Say

Welcome to Unit: Quality Control and Testing of Biomass Pellets. Biomass pellets are only as good as their quality—whether they burn well, store safely, or meet standards depends on how we test them. Today, we'll dive into testing protocols for key factors like energy value, ash, and moisture, and get hands-on with a moisture meter to check agricultural waste. By the end, you'll know how to ensure top-notch pellets for bioenergy. Let's test and perfect!

### Ask

Ask the participants

What makes a biomass pellet 'good' or 'bad' for burning?

Why might too much water in pellets be a problem?

Have you ever tested something—like food or soil—for quality? How?

### Notes for Facilitation

- Encourage quick responses (e.g., "Burns hot," "Won't light," "Dried it").
- Note ideas (e.g., "energy," "moisture," "check") on a whiteboard.
- Transition: "These are key quality clues—let's see how we measure them precisely.
- Use visuals (e.g., calorimeter photo, moisture meter types) on slides or handouts.
- Pause for quick Q&A (e.g., "Why test ash?").
- Relate to local waste (e.g., corn stalks, sawdust).
- Provide a checklist (e.g., "Calibrate, measure, interpret 10-15%").
- Offer feedback: "Good probe placement—mention GCV next time!"
- Debrief: "What was trickiest to measure?"

## Explain

### Standard Protocol for Testing Pellets and Important Parameters (30 minutes)

#### Key Points:

- **Gross Calorific Value (GCV):**
  - Measures energy (kcal/kg or MJ/kg).
  - Test: Bomb calorimeter—ignite 1g, measure temp rise (ASTM D5865).
  - Why: High GCV = more energy.
- **Ash Content:**
  - Residue after burning (%).
  - Test: Muffle furnace—burn 1g at 550-600°C, weigh ash (ASTM E1755).
  - Why: Low ash = cleaner burn.
- **Moisture Content:**
  - Water in pellets (%).
  - Test: Drying oven—dry 10g at 105°C, weigh loss (ASTM E871).
  - Why: 10-15% = efficient burn.

#### Example:

“High ash from dirty biomass clogs a boiler—testing keeps it low for smooth burning.”

### 7.3.2 Use of Moisture Meter for Agricultural Waste (30 minutes)

#### Key Points:

- **Types:** Pin-type (probes), pinless (sensors), infrared (surface).
- **Steps:**
  1. Prep sample (clean, uniform).
  2. Calibrate meter (set for biomass).
  3. Measure (insert probes or scan).
  4. Interpret (10-15% ideal).
- **Why:** Right moisture prevents weak pellets or clogs.

#### Example:

“Rice husk at 20% moisture? Too wet—dry it down to 12% for perfect pelleting.”

## Role Play

#### Activity:

- **Setup:** Pair participants—one as tester, one as quality checker.
- **Task:** Tester uses a mock moisture meter on waste (e.g., straw, wood chips) and explains GCV/ash/moisture importance; checker ensures correct steps and results.
- **Scenarios:**
  1. Test sawdust—aim for 12% moisture.
  2. Check corn stalks—adjust if over 15%.
- Switch roles after 10 minutes.

## Summarize

Summarize the topics covered in the module.

- Today, we've learned how to test pellets for GCV, ash, and moisture using standard protocols—key to energy and quality. We've also mastered the moisture meter to keep agricultural waste in the 10-15% sweet spot. These skills ensure top pellets for bioenergy. Any final questions?
- You may ask the learners to recall what all has been taught.



## UNIT 7.4: Post-Production Maintenance and Best Practices in Bioenergy Plants

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Perform the post-production cleaning and maintenance process followed in the industry.
2. Explain Do's and Don'ts in biomass pellet manufacturing.

### Say

- Welcome to Unit 7.4: Post-Production Maintenance and Best Practices in Bioenergy Plants. After making biomass pellets, the job isn't over—cleaning up and maintaining machines keeps everything running smoothly and safely. Today, we'll master the industry's cleaning and maintenance steps and learn the key Do's and Don'ts of pellet production. These skills ensure quality, cut downtime, and keep the plant safe. Let's roll up our sleeves and get started!

### Ask

Ask the participants

What happens if you don't clean a machine after using it?"

What's one safety rule you'd set for a pellet plant?"

Have you ever fixed or cleaned equipment? What did you do?

### Notes for Facilitation

- Encourage quick answers (e.g., "Clogs up," "Wear gloves," "Wiped it down").
- Note ideas (e.g., "dust," "safety," "oil") on a whiteboard.
- Transition: "These are great instincts—let's build on them with pro practices."

### Role Play

**Activity:**

**Setup:** Pair participants—one as technician, one as supervisor.

**Task:** Technician cleans and maintains a mock machine (e.g., conveyor, mill), citing 2 Do's; supervisor checks for 2 Don'ts avoided.

**Scenarios:**

1. Clean a dusty dryer—lubricate, check filters.
2. Maintain a pellet mill—clear die, tension belts.

Switch roles after 10 minutes.

## Explain

### Cleaning and Maintenance of Machines After Production

#### Key Points:

- **Importance:** Prevents contamination, extends machine life, ensures safety.
- **Cleaning:**
  - Shutdown: Power off, LOTO, cool down.
  - Tools: Air, vacuums, brushes, water jets.
  - Areas: Pellet mill (die/rollers), conveyors, dryers, dust collectors, coolers.
  - Waste: Segregate, dispose per regs.
- **Maintenance:**
  - Routine: Check wear, lubricate, tension belts.
  - Preventive: Schedule oil/filter changes, inspect systems.
  - Corrective: Fix faults, log actions.
- **Safety:** LOTO, PPE, ventilate, keep fire gear handy.

#### Example:

“Pellet mill die clogged with sawdust? Scrape it clean after shutdown—keeps pellets pure.”

### Do's and Don'ts in Biomass Pellet Manufacturing

#### Key Points:

- **Do's:**
  - Use clean biomass, control moisture (10-15%), follow manuals, test quality, maintain emission controls, wear PPE, train staff.
- **Don'ts:**
  - Don't use wet/contaminated material, overload machines, skip safety checks, ignore noise, store damp pellets, neglect fire risks.

#### Example:

“Do check moisture with a meter; don't overload the mill—it jams and risks a fire.”

## Summarize

Summarize the topics covered in the module.

Today, we've learned how to clean and maintain machines post-production—shutdown, scrub, lube, and log it. We've also nailed the Do's and Don'ts for safe, quality pellet manufacturing. These practices keep plants running and risks low. Any final thoughts?

You may ask the learners to recall what all has been taught.



## 8. Ensure installation and operation of small biogas plant

Unit 8.1: Introduction to Biogas and Comparative Analysis

Unit 8.2: Site Assessment and Material Selection for Biogas Plant Construction

Unit 8.3: Construction and Installation of Biogas Plant

Unit 8.4: Operation, Maintenance, and Safety Practices

Unit 8.5: Utilization of Biogas and Bio-Slurry for Sustainable Benefits



## Key Learning Outcomes



**At the end of this module, you will be able to:**

1. Understanding Biogas Basics and differentiate it from LPG and CNG.
2. Site and Material Assessment for biogas plant installation based on terrain, water availability, and distance from the kitchen.
3. Identify key components required for constructing a biogas plant.
4. Select appropriate types of Indian biogas digesters based on design and agro-climatic conditions.
5. Perform Installation and Fixing of Install inlet and outlet pipes, and assemble prefabricated or modular biogas plants.
6. Know the Plant Operation and Routine maintenance practices for gas pipelines, burners, and appliances.
7. Learn safety protocols and the dos and don'ts of biogas plant usage.
8. Understand the role of bio slurry as a nutrient-rich fertilizer.
9. Understand benefits of biogas at the community level.
10. Promote biogas as a sustainable and green alternative to traditional fuels.

## UNIT 8.1: Introduction to Biogas and Comparative Analysis

### Unit Objectives

At the end of this unit, you will be able to:

1. Discuss overview of biogas technology
2. Identify and list out Difference between biogas, LPG, and CNG
3. Discuss environmental and economic benefits of biogas

### Say

Welcome to Unit Today, we'll explore biogas technology, a renewable energy source that's transforming waste into clean energy. We'll also compare it with LPG and CNG and discuss its environmental and economic benefits.

### Ask

Ask the participants

- Has anyone heard of biogas before? What do you think it is used for?
- Why might renewable energy sources like biogas be important in today's world?

### Notes for Facilitation

- Encourage brief responses to gauge prior knowledge.
- Keep the tone engaging and curious to spark interest.
- Ensure groups have access to the text or a summary handout.
- Encourage creativity in presentations (e.g., drawings or role-play snippets).
- Assign roles based on learners' interests or randomly.
- Guide the role play to focus on benefits from the text.
- Provide paper, markers, or digital tools for sketching.
- Encourage practical ideas based on local waste sources (e.g., food scraps, manure).

## Explain

1. Define biogas: "Biogas is a renewable energy source produced by anaerobic digestion of organic waste like manure, food scraps, and agricultural residues. It's mainly methane and carbon dioxide, making it a clean fuel for cooking, heating, or electricity."
2. Outline the process: "It happens in a biogas plant, where organic waste is broken down by microorganisms in a digester. The result is biogas and a by-product called digestate, which can be used as fertilizer."
3. Highlight advantages: "It's sustainable, reduces waste, and cuts greenhouse gas emissions. Let's compare biogas with LPG and CNG. All three are fuels, but they differ in source, composition, and use."

Use Table 13 from the text to summarize:

- **Biogas:** Renewable, 50-70% methane, from organic waste, used for cooking and electricity.
- **LPG:** Fossil fuel-based, propane and butane, for cooking and heating.
- **CNG:** Mostly methane, from natural gas, used in vehicles and power generation.

Highlight key differences: "Biogas is renewable and eco-friendly, while LPG and CNG come from fossil fuels. Biogas also produces fertilizer as a bonus!"

**Environmental Benefits:** Biogas reduces methane emissions—a potent greenhouse gas—by capturing it from waste. It also cuts landfill use and provides organic fertilizer, supporting sustainable farming.

**Economic Benefits:** It creates jobs, saves waste management costs, and reduces reliance on imported fuels. Farmers save money with free fertilizer, and plants can earn carbon credits.

Emphasize: "Biogas is a win-win: good for the planet and the economy."

## Team Activity

Divide the class into five to six groups of four participants each.

**Activity:** "Fuel Comparison Chart"

Divide learners into small groups.

Provide a blank table with columns: Source, Composition, Production, Applications, Pros, Cons.

Assign each group one fuel (biogas, LPG, or CNG) to fill in based on the text.

Groups present their findings to the class.

**Duration:** 15-20 minutes.

## Demonstrate

Show a video

### VIDEO:

Topic: Making a Biogas Plant

Link: <https://www.youtube.com/watch?v=GoVcFFQW5UY>

Duration: 15 m 01 sec



Scan Qr to watch video

## Summarize

Summarize the topics covered in the module.

- Today, we learned that biogas is a renewable fuel made from organic waste through anaerobic digestion. It's different from LPG and CNG because it's sustainable and produces fertilizer. Its benefits include reducing emissions, managing waste, creating jobs, and supporting energy independence.
- You may ask the learners to recall what all has been taught.



## UNIT 8.2: Site Assessment and Material Selection for Biogas Plant Construction

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Identify and list components for biogas plant construction
2. Assess site suitability for biogas plant installation
3. Determine plant size and material requirements for Bio gas Plant
4. Evaluate the feasibility of raw material sourcing
5. Analyse household energy needs for plant capacity planning

### Say

Welcome to Unit ! Today, we'll dive into the practical steps of planning a biogas plant. We'll look at what components you need, how to pick the right site, size the plant, source materials, and match it to household energy needs.

### Ask

Ask the participants

- What do you think makes a good location for a biogas plant?
- Why might choosing the right materials matter for a biogas plant?
- Which component do you think is most critical? Why?
- What materials do you see around you that could be used for a biogas plant?
- What's one site factor you'd prioritize? Why?
- How might weather affect your site choice?
- How would you adjust plant size if feedstock doubled?
- What local materials could you use instead of steel?
- What waste sources are common where you live?
- How could seasonal changes affect your plan?
- How would cooking habits change your plant size?
- Which digester type would you pick for your home? Why?

## Notes for Facilitation

- Keep the intro short and engaging to set the stage.
- Use learner responses to transition into the first topic.
- Use visuals to make abstract parts concrete.
- Encourage learners to think locally about materials.
- Provide a handout with site factors from the text.
- Guide groups to consider trade-offs (e.g., cost vs. convenience).
- Simplify math for clarity; encourage calculator use.
- Relate materials to learners' environments.
- Provide a template with the formula and digester options.
- Encourage practical, context-based choices.

## Demonstrate

Show a labeled diagram (e.g., Figure 55 from the text) or a simple model of a biogas plant. Point out the digester, gas holder, inlet, and outlet.

If possible, use a short video clip or animation to show how components work together.

Walk through the example calculation from the text on a whiteboard or screen:  $V = (40 \text{ kg/day} \times 1.5 \text{ L/kg} \times 40 \text{ days}) / 0.8 = 3 \text{ m}^3$ .

Show samples or images of materials (e.g., PVC pipe, concrete block) if available.

## Role Play

- **Scenario:** "Planning a Biogas Supply Chain"
  - Roles: Farmer (supplies dung), Transporter (moves waste), Plant Manager (assesses needs), Community Leader (approves site).
  - Task: Negotiate how to source and deliver raw materials for a plant.
  - Duration: 10-15 minutes.
- Debrief: "What sourcing challenges came up? How did you solve them?"

## Summarize

Summarize the topics covered in the module.

- Today, we covered the nuts and bolts of biogas plant planning. We listed components like digesters and pipelines, assessed sites for suitability, sized plants with feedstock and energy needs, sourced raw materials, and matched capacity to households. You're now ready to plan a real biogas plant!

## UNIT 8.3: Construction and Installation of Biogas Plant

### Unit Objectives



**At the end of this unit, you will be able to:**

1. Prepare the site by conducting layout marking, excavation, leveling, and bed concrete casting for biogas plant installation.
2. Fabricate, assemble, and install the digester tank, outlet chamber, inlet tank, displacement chamber, and gas holder as per technical specifications.
3. Install and lay out the gas distribution pipeline, connect the gas regulator and stove, and demonstrate proper lighting and usage of the biogas stove.
4. Identify and arrange construction materials, estimate manpower requirements, ensure curing of civil works, and provide guidelines on the safe usage of biogas systems.

### Say



Welcome to the session on site preparation for biogas plant installation. This is a crucial step that ensures the structural stability and longevity of the plant.

In first session, we will learn about the fabrication, assembly, and installation of the core components of a biogas plant.

Next, we will focus on the installation of gas pipelines, regulators, and stoves. Proper installation ensures a safe and efficient gas supply

In the final session, we will discuss construction material management, manpower estimation, and safety guidelines.

### Ask



Ask the participants

- Why do you think site selection and preparation are important for biogas plant installation?
- Can you name some key components of a biogas plant and their functions?
- What precautions would you take when installing gas pipelines and connecting stoves?
- Why is it important to estimate material requirements accurately?

### Notes for Facilitation



- Share real-world examples of site failures due to poor preparation.
- Provide handouts of site layout diagrams and photos.
- Provide physical samples or visual aids of construction materials.
- Display videos of fabrication and installation processes.
- Show a sample pipeline setup using diagrams and actual components.
- Highlight safety guidelines through a visual presentation.
- Provide sample material lists and resource planning templates.
- Share case studies on resource mismanagement and its impact.

## Explain

- Discuss the importance of layout marking, excavation, leveling, and bed concrete casting.
- Emphasize the need for accurate site measurement and safety precautions.
- Discuss the purpose of the digester tank, inlet tank, displacement chamber, outlet chamber, and gas holder.
- Explain the material selection criteria and construction process.
- Discuss the purpose of the digester tank, inlet tank, displacement chamber, outlet chamber, and gas holder.
- Explain the material selection criteria and construction process.
- Discuss the purpose of the digester tank, inlet tank, displacement chamber, outlet chamber, and gas holder.
- Explain the material selection criteria and construction process.

## Demonstrate

- Conduct a practical demonstration of layout marking using lime powder, pegs, and measuring tools.
- Show the correct excavation depth and how to ensure even levelling using a spirit level.
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- Conduct a practical demonstration of layout marking using lime powder, pegs, and measuring tools.
- Show the correct excavation depth and how to ensure even levelling using a spirit level.
- Provide sample material lists and resource planning templates.
- Share case studies on resource mismanagement and its impact.

## Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Form teams to practice pipeline assembly and gas stove connection using training models.

Give the groups 15 minutes for the activity.

## Summarize

Summarize the topics covered in the module.

- Recap key points on gas pipeline installation and safety.
  - Encourage participants to share their observations.
  - Conclude with a checklist for material management and safety compliance.
  - Reinforce the importance of continuous monitoring and user education.
- Provide feedback and clarify any remaining questions.

## UNIT 8.4: Operation, Maintenance, and Safety Practices

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Conduct gas leakage tests and ensure safe operational procedures during biogas plant operation.
2. Demonstrate the proper charging of the biogas plant with the correct ratio of cattle dung and water, ensuring efficient feedstock preparation and input.
3. Educate users on the dos and don'ts of biogas plant operation, including regular input management, slurry collection, and maintenance practices.
4. Perform routine maintenance checks on the gas pipeline, plant components, and associated apparatus to ensure safety and optimal performance.

### Say

Welcome to Unit 8.4! Today, we'll focus on keeping a biogas plant running safely and efficiently. We'll learn how to test for gas leaks, charge the plant correctly, follow key dos and don'ts, and maintain it for the long haul.

### Ask

Ask the participants

- What do you think could go wrong if a biogas plant isn't maintained properly?
- Why might safety be a big deal when working with biogas?
- What would you do if you found a leak during a test?
- Why is ventilation so important around a biogas plant?
- What might happen if you use too much water or dung?
- How could you tell if the slurry isn't mixed well?
- What's one 'do' you'd always follow? Why?
- What's a 'don't' that could cause big trouble?
- What's one maintenance task you'd prioritize?
- How could skipping maintenance affect gas output?

### Notes for Facilitation

- Use responses to highlight the importance of safety and maintenance.
- Set an engaging tone to emphasize practical skills.
- Keep the demo simple and safe—no real gas needed.
- Emphasize immediate action for leaks to build confidence.
- Use safe substitutes for dung to keep it hands-on.

## Explain

- Gas leaks are dangerous—they can cause fires or explosions. Regular tests help catch them early. You'll use tools like soap solution or a gas detector to spot leaks in pipelines and valves.
- Procedure: Start with a visual check, then apply soap solution—bubbles mean a leak. Use a detector for precision, monitor pressure, and fix issues right away.
- Safety: Ventilate the area, install flame arrestors, and train everyone on emergency steps like shutting off the gas.
- Charging the plant right keeps gas flowing. The ideal mix is 1:1 cattle dung to water by weight—like 100 kg dung with 100 liters water. This slurry feeds the microbes that make biogas.
- Steps: Collect fresh dung, mix it with water in a pit, stir it smooth, and pour it into the inlet. Watch for spills or clogs.
- Tips: Stick to a daily schedule and avoid overfeeding to prevent problems like foaming or low gas output.
- Following simple rules keeps the plant safe and productive. Do check for leaks, feed it daily, and clean the chambers. Don't overfeed, add trash like plastic, or ignore small leaks.
- Examples: Regular slurry input keeps gas steady. Skipping maintenance can clog the system or worse.
- Maintenance stops breakdowns and keeps the plant safe. Check pipelines for rust, the digester for leaks, and valves for function. Clean sludge, lubricate parts, and monitor pressure.
- Steps: Look at everything, test for leaks, clean, fix what's broken. Do this weekly, with a big check monthly.  
Benefits: "It saves money on repairs and keeps gas flowing."

## Demonstrate

- Simulate a gas leakage test: Use a small pipe or balloon with air, apply soap solution, and show bubbles forming. Explain how this applies to a real biogas plant.
- Show a pressure gauge (or image) and explain normal vs. irregular readings.  
Show the video

### VIDEO

Topic: Mastering Preventive Maintenance:  
Step-by-Step Guide to Crafting an Effective Plan  
By: Limble CMMS  
Link: <https://www.youtube.com/watch?v=GAQkuNEHCZA>  
Duration: 03 m 59 sec



Scan Qr to watch video

## Team Activity



**Activity:** "Maintenance Inspection"

- Groups get a checklist (pipelines, digester, valves, etc.) and a mock plant scenario (e.g., "rusty pipe, low pressure").
- Task: Identify issues, propose fixes, and explain why.
- Duration: 20-25 minutes.

Share findings with the class.

## Summarize



Summarize the topics covered in the module.

Today, we learned how to keep a biogas plant safe and running smoothly. We tested for leaks, charged it with the right slurry mix, followed dos and don'ts, and checked components regularly. You're now ready to operate and maintain a plant like a pro!  
Provide feedback and clarify any remaining questions.



## UNIT 8.5: Utilization of Biogas and Bio-Slurry for Sustainable Benefits

### Unit Objectives

**At the end of this unit, you will be able to:**

1. Identify and assess the feasibility of methane utilization as a green substitute for LPG, wood, cattle dung, and kerosene through biogas plants, promoting sustainable energy solutions with active community participation.
2. Perform the correct procedures for drying bio slurry, ensuring proper moisture control and preventing nutrient loss, followed by effective packaging for storage and distribution as compost.
3. Demonstrate the effective utilization of bio slurry as an organic fertilizer in agricultural fields and as a nutrient source in fishery ponds, ensuring appropriate application techniques for maximum productivity.

### Say

Welcome to Unit 8.5! Today, we're diving into the practical uses of biogas and bio-slurry. We'll explore how methane can replace traditional fuels, how to dry and package slurry for compost, and how to apply it in farming and fisheries for sustainable gains.

### Ask

Ask the participants

- What fuels do people in your area rely on? How might biogas change that?
- What do you think bio-slurry could do for crops or fish ponds near you?
- Which fuel in your community could biogas replace most easily? Why?
- How could you convince people to switch to biogas?
- What might happen if slurry isn't dried enough?"
- Would sun or mechanical drying work better where you are? Why?

### Notes for Facilitation

- Use responses to link the unit to learners' contexts.
- Set a hands-on, solution-oriented tone.
- Keep math simple; focus on the impact.
- Relate to local fuel use patterns.
- What might happen if slurry isn't dried enough?
- Would sun or mechanical drying work better where you are? Why?
- Provide a template with application steps.
- Encourage local crop or fish examples.

## Explain

- Methane from biogas is a renewable fuel that can swap out LPG, wood, dung, and kerosene. It cuts emissions—like saving 144 kg of CO<sub>2</sub> monthly by replacing 50 liters of LPG—and reduces waste.
- Feasibility check: "Look at feedstock supply, plant costs, and savings. Community involvement—through training and cooperatives—makes it work.  
Benefits: "Cleaner air, less deforestation, and local jobs."
- Bio-slurry needs drying to 10-15% moisture for compost. Sun drying—spread it thin, turn it—works for small setups. Mechanical drying is faster but pricier."  
Nutrient care: "Dry at 60-70°C and use shade to keep nutrients like nitrogen intact.  
Packaging: "Cool it, bag it in 25 kg sacks, seal, and label for storage or sale."
- Bio-slurry is a nutrient powerhouse. For crops, dilute it 1:1 and mix into soil or spray it 1:5 on leaves—it boosts growth and soil health.
- For fish ponds: "Dilute 1:3, apply weekly—it feeds plankton for fish. Monitor water to avoid overdoing it."
- Tips: "Time it right—early morning for crops, regular checks for ponds."

## Demonstrate

- Show the emission reduction calculation on a whiteboard: LPG (50 L × 2.983 kg CO<sub>2</sub>/L = 149.15 kg) vs. biogas (50 m<sup>3</sup> × 0.1 kg CO<sub>2</sub>/m<sup>3</sup> = 5 kg), netting 144.15 kg saved.
- Display a biogas stove (or photo) to show its practical use.
- Mimic sun drying: Spread wet soil or dough (2-3 cm thick) on a tray, "turn" it with a spoon, and show a dry sample (e.g., crumbled soil).
- Simulate packaging: Fill a small bag with dry material, seal it, and add a mock label (e.g., "Compost, 15% moisture, March 2025").
- Show a short clip or diagram of slurry use (e.g., spreading on a field or adding to a pond) to reinforce techniques.

## Team Activity

- Divide the class into five to six groups of four participants each.

**Activity:** "Slurry Application Strategy"

Groups get a case (e.g., wheat farm, carp pond).

Task: Plan slurry use—how much, how to apply, when—and list benefits.

Duration: 20-25 minutes.

Present plans to the class.

## Summarize

Summarize the topics covered in the module.

Today, we unlocked the power of biogas and bio-slurry. We saw how methane can replace dirty fuels, learned to dry and package slurry for compost, and applied it to boost farming and fisheries. You're ready to make sustainability real!

Provide feedback and clarify any remaining questions.





## 6. Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

[Employability skills workbook](#)



Scan/ Click QR Code to access







| Module No.                                                                           | Unit No. | Topic Name                                                                     | URL                                                                                                                         | QR Code(s)                                                                            |
|--------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Module 1 -</b><br>Entrepreneurship and Basics of Biomass Energy                   | Unit 1.1 | NREL Energy Basics: Biomass.                                                   | <a href="https://www.youtube.com/watch?v=7cCSV0IO4zE&amp;t=193s">https://www.youtube.com/watch?v=7cCSV0IO4zE&amp;t=193s</a> |    |
| <b>Module 2-</b><br>Manage financial budget and developing business plans            | Unit 2.1 | Break-Even Analysis with Easy Explanation                                      | <a href="https://www.youtube.com/watch?v=uKeopB8acGQ">https://www.youtube.com/watch?v=uKeopB8acGQ</a>                       |    |
| <b>Module 2-</b><br>Manage financial budget and developing business plans            | Unit 2.2 | Sources of finance                                                             | <a href="https://www.youtube.com/watch?v=Jj0LJQlee14">https://www.youtube.com/watch?v=Jj0LJQlee14</a>                       |   |
| <b>Module 3-</b><br>Manage sales, supply, and marketing of product                   | Unit 3.3 | How To Measure Marketing Performance                                           | <a href="https://www.youtube.com/sorts/wp5B-YGb1UU">https://www.youtube.com/sorts/wp5B-YGb1UU</a>                           |  |
| <b>Module 4-</b><br>Assess Compliance with Applicable Laws, Policies, and Procedures | Unit 4.2 | CTE / CTO Consent to Established / Operate, CPCC / CPCB, SPCB, Pollution index | <a href="https://www.youtube.com/watch?v=3Wh5pwNHt7E">https://www.youtube.com/watch?v=3Wh5pwNHt7E</a>                       |  |
| <b>Module 4-</b><br>Assess Compliance with Applicable Laws, Policies, and Procedures | Unit 4.3 | How to Register a Company in India: Business Structure, Documentation,         | <a href="https://www.youtube.com/watch?v=KjsFMOlxQYk">https://www.youtube.com/watch?v=KjsFMOlxQYk</a>                       |  |



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|--------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Module 4-</b><br>Assess<br>Compliance<br>with Applicable<br>Laws, Policies,<br>and Procedures | Unit<br>4.4 | Complete<br>Government<br>Schemes Course<br>–<br>GOBARDhan<br>scheme, SATAT<br>Scheme,<br>National<br>Bioenergy<br>Programme | <a href="https://www.youtube.com/watch?v=cFtU0xWjqyw">https://www.youtube.com/watch?v=cFtU0xWjqyw</a>                                 |    |
| <b>Module 5-</b><br>Health & Safety<br>in Bioenergy<br>Manufacturing<br>facility                 | Unit<br>5.2 | Administering<br>First Aid                                                                                                   | <a href="https://www.youtube.com/watch?v=7TSqfYfdlb8">https://www.youtube.com/watch?v=7TSqfYfdlb8</a>                                 |    |
| <b>Module 5-</b><br>Health & Safety<br>in Bioenergy<br>Manufacturing<br>facility                 | Unit<br>5.3 | Personal<br>Protection<br>Equipment                                                                                          | <a href="https://www.youtube.com/watch?v=YE-63aeECVs">https://www.youtube.com/watch?v=YE-63aeECVs</a>                                 |   |
| <b>Module 7-</b><br>Ensure<br>Manufacturing<br>of Biomass<br>pellet.                             | Unit<br>7.1 | Flat Die Biomass<br>Pellet Making<br>Plant #Pellet<br>#woodpelletmac<br>hine                                                 | <a href="https://www.youtube.com/shorts/jU47s7zDdgQ">https://www.youtube.com/shorts/jU47s7zDdgQ</a>                                   |  |
| <b>Module 7-</b><br>Ensure<br>Manufacturing<br>of Biomass<br>pellet.                             | Unit<br>7.1 | Pellet Cooling<br>Process                                                                                                    | <a href="https://youtube.com/shorts/vcyB_hLkqwQ?si=JaMa17zDL_LlL4Ogg">https://youtube.com/shorts/vcyB_hLkqwQ?si=JaMa17zDL_LlL4Ogg</a> |  |
| <b>Module 7-</b><br>Ensure<br>Manufacturing<br>of Biomass<br>pellet.                             | Unit<br>7.1 | The Complete<br>Pelletizing<br>Process of<br>Biomass/Wood<br>Pellet Mill                                                     | <a href="https://www.youtube.com/watch?v=ON6pl4pjpV0">https://www.youtube.com/watch?v=ON6pl4pjpV0</a>                                 |  |

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|-----------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Module 8-</b><br>Ensure<br>installation and<br>operation of<br>small biogas<br>plant | Unit<br>8.1 | Making a Biogas<br>Plant                                                                          | <a href="https://www.youtube.com/watch?v=GoVcFFQW5UY">https://www.youtube.com/<br/>watch?v=GoVcFFQW5UY</a> |  |
| <b>Module 8-</b><br>Ensure<br>installation and<br>operation of<br>small biogas<br>plant | Unit<br>8.4 | Mastering<br>Preventive<br>Maintenance:<br>Step-by-Step<br>Guide to Crafting<br>an Effective Plan | <a href="https://www.youtube.com/watch?v=GAQkuNEHCZA">https://www.youtube.com/<br/>watch?v=GAQkuNEHCZA</a> |  |



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